



NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

STR

LAB #1

SW INSTALLATION & INTERACTION WITH THE “PHYSICAL” SYSTEM

2022 - 2023

PART 1:

Software installation & configuration

- ☐ Software installation
- ☐ Creation and configuration of new project for windows 32 console application
- ☐ Adding components to the project

PART 2:

Interaction with physical system

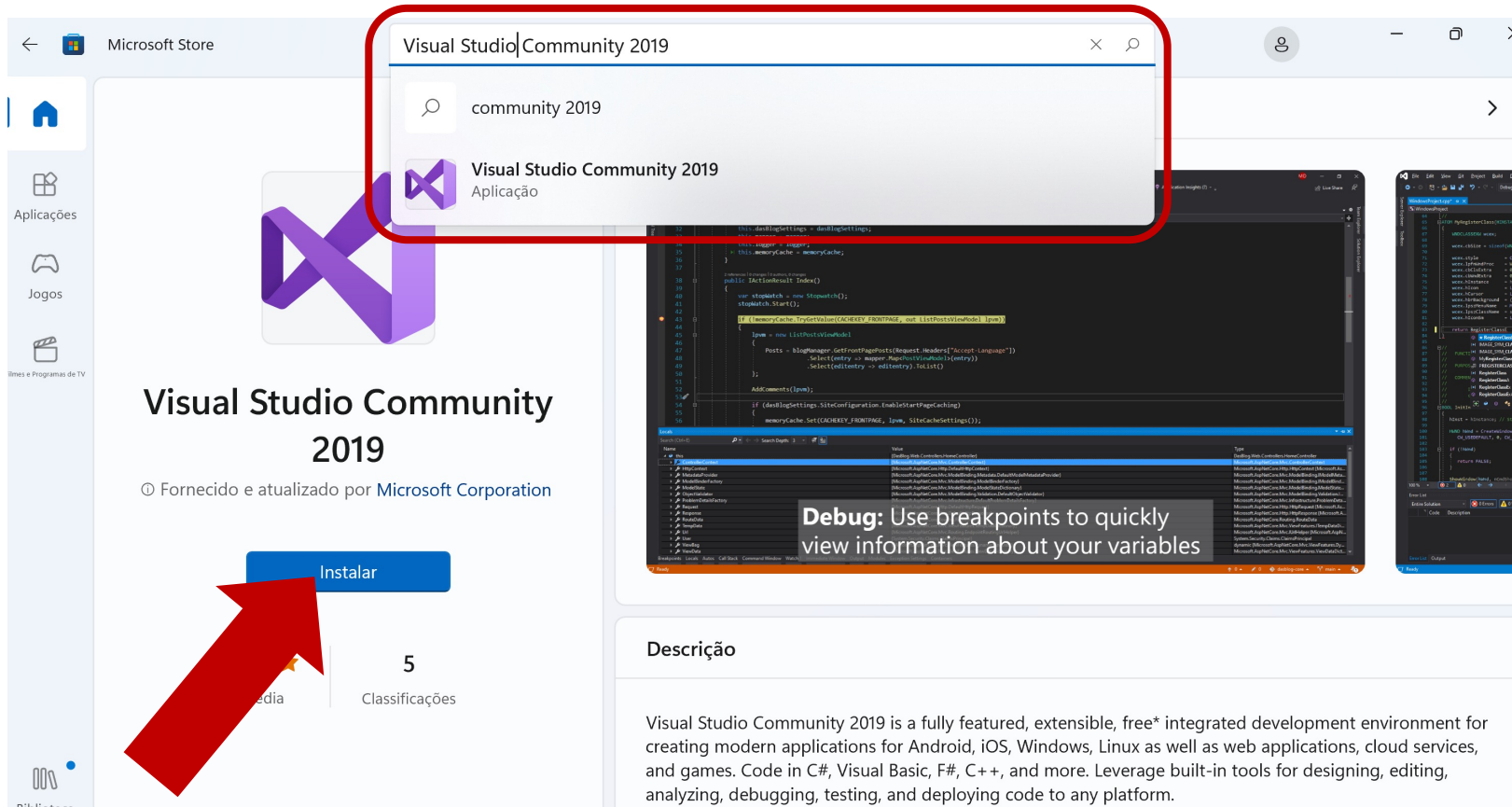
- ☐ Using the Digital Acquisition board (DAQ)
- ☐ Introducing the kit “compact storage”
- ☐ The physical vs simulated storage kit
- ☐ Reading sensors & using the actuators

PART 1

Software installation & configuration

INSTALLATION OF VISUAL STUDIO

Install Visual Studio Community 2019 using Microsoft Store



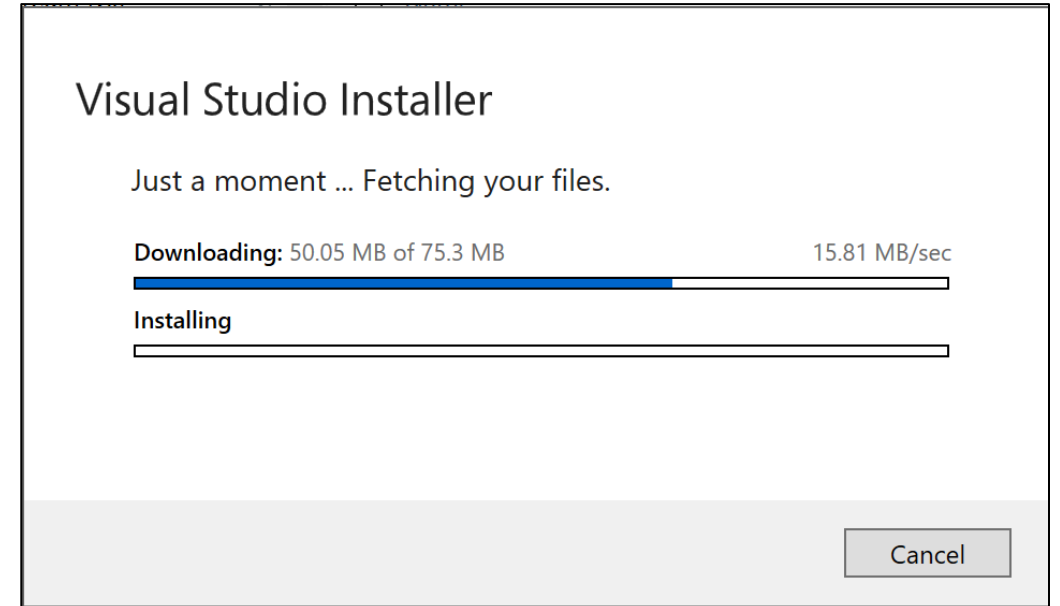
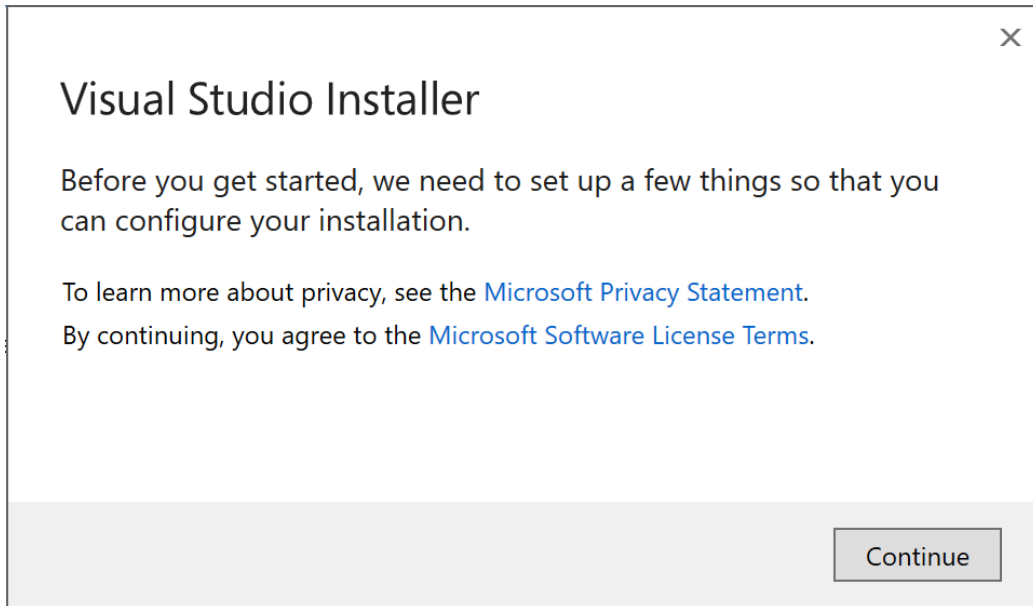
Recommended version:

- Visual Studio 2019 (VS19)
- Community edition

INSTALLATION OF VISUAL STUDIO



Run installation file



INSTALLATION OF VISUAL STUDIO



VS Community 2019 features configuration

Installing — Visual Studio Community 2019 — 16.11.3

Workloads Individual components Language packs Installation locations

Need help choosing what to install? [More info](#)

Web & Cloud (4)

- ☐ ASP.NET and web development
Build web applications using ASP.NET Core, ASP.NET, HTML/JavaScript, and Containers including Docker supp...
- ☐ Python development
Editing, debugging, interactive development and source control for Python.
- ☐ Azure development
Azure SDKs, tools, and projects for developing cloud apps and creating resources using .NET and .NET Framework...
- ☐ Node.js development
Build scalable network applications using Node.js, an asynchronous event-driven JavaScript runtime.

Desktop & Mobile (5)

- ☐ .NET desktop development
Build WPF, Windows Forms, and console applications using C#, Visual Basic, and F# with .NET and .NET Frame...
- ☒ Desktop development with C++
Build modern C++ apps for Windows using tools of your choice, including MSVC, Clang, CMake, or MSBuild.
- ☐ Universal Windows Platform development
Create applications for the Universal Windows Platform with C#, VB, or optionally C++.
- ☐ Mobile development with .NET
Build cross-platform applications for iOS, Android or Windows using Xamarin.
- ☐ Mobile development with C++
Build cross-platform applications for iOS, Android or Windows using C++.

Gaming (2)

- ☐ Game development with Unity
Create 2D and 3D games with Unity, a powerful cross-platform development environment.
- ☐ Game development with C++
Use the full power of C++ to build professional games powered by DirectX, Unreal, or Cocos2d.

Location
C:\Program Files (x86)\Microsoft Visual Studio\2019\Community [Change...](#)

By continuing, you agree to the [license](#) for the Visual Studio edition you selected. We also offer the ability to download other software with Visual Studio. This software is licensed separately, as set out in the [3rd Party Notices](#) or in its accompanying license. By continuing, you also agree to those licenses.

Installation details

- Visual Studio core editor
- Desktop development with C++
 - Included
 - ✓ C++ core desktop features
 - Optional
 - ✓ MSVC v142 - VS 2019 C++ x64/x86 build tools
 - ✓ Windows 10 SDK (10.0.19041.0)
 - ✓ Just-In-Time debugger
 - ✓ C++ profiling tools
 - ✓ C++ CMake tools for Windows
 - ✓ C++ ATL for latest v142 build tools (x64/x86 &...)
 - ✓ Test Adapter for Boost.Test
 - ✓ Test Adapter for Google Test
 - ✓ Live Share
 - ✓ IntelliCode
 - ✓ C++ AddressSanitizer
 - ☐ MSVC v142 - VS 2019 C++ ARM64 build tools
 - ☐ C++ MFC for latest v142 build tools (x86...
 - ☐ C++/CLI support for v142 build tools (Late...
 - ☐ C++ Modules for v142 build tools (x64/x8...
 - ☐ C++ Clang tools for Windows (12.0.0 - x64...
 - ☐ JavaScript diagnostics
 - ☐ Incredibuild - Build Acceleration
 - ☐ Windows 10 SDK (10.0.18362.0)
 - ☐ Windows 10 SDK (10.0.17763.0)
 - ☐ Windows 10 SDK (10.0.17134.0)
 - ☐ Windows 10 SDK (10.0.16299.0)
 - ☐ MSVC v141 - VS 2017 C++ x64/x86 build t...
 - ☐ MSVC v140 - VS 2015 C++ build tools (v1...

Total space required 8.03 GB

☐ Install while downloading ☒ Install

- To avoid taking unnecessary disk space, select at least these features.
- Additional features maybe installed latter on if necessary.

INSTALLATION OF VISUAL STUDIO



Visual Studio Installer

Installed Available

Visual Studio Community 2019

Downloaded

Installing: package 374 of 404

61%

Win10SDK_10.0.19041

☐ Start after installation

[Release notes](#)

Pause

Developer News

[Visual Studio 2022 Preview 4 is now available](#)
We are excited to announce the fourth preview re...
Thursday, September 16, 2021

[Announcing .NET 6 Release Candidate 1](#)
We are happy to release .NET 6 Release Candidat...
Thursday, September 16, 2021

[Boost your productivity with Productivity Power Tools Extensions in Visual Studio 2022](#)
Visual Studio 2022 is here and is more customiza...
Thursday, September 16, 2021

[View more Microsoft developer news...](#)

Need help? Check out the [Microsoft Developer Community](#) or reach us via [Visual Studio Support](#).

Installer version 2.11.40.25675

Wait till installation is complete

INSTALLATION OF VISUAL STUDIO



Visual Studio Installer

Installed Available



Visual Studio Community 2019

16.11.3

⚠ A restart is required. If needed, any remainin

Restart

Reboot required

Success! One more step to go. Please restart your computer before you start Visual S

Restart

Cancel

Developer News

[Visual Studio 2022 Preview 4 is now available](#)

We are excited to announce the fourth preview re...
Thursday, September 16, 2021

[Announcing .NET 6 Release Candidate 1](#)

We are happy to release .NET 6 Release Candidat...
Thursday, September 16, 2021

[Boost your productivity with Productivity Power Tools Extensions in Visual Studio 2022](#)

Visual Studio 2022 is here and is more customiza...
Thursday, September 16, 2021

[View more Microsoft developer news...](#)

Need help? Check out the [Microsoft Developer Community](#) or reach us via [Visual Studio Support](#).

Installer version 2.11.40.25675

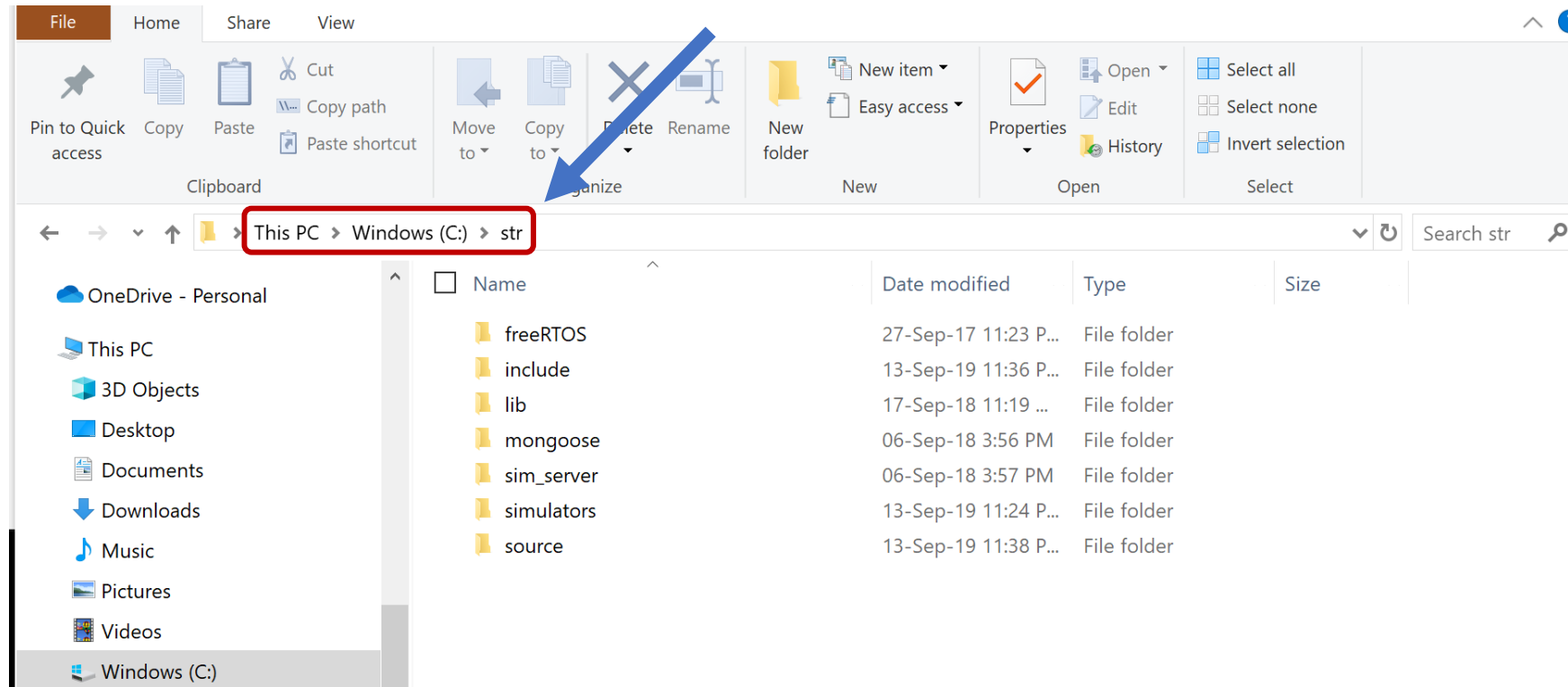
Reboot or proceed

- Depending on your Windows configuration, either you are asked to reboot or take to the login GUI.
 - Case the login shows up, see how to do it in the “signing in” Slide.

LABWORK1 PROJECT PACKAGE



- ❑ Copy the material for Lab work from CLIP
- ❑ Extract **STR_TP1_storage.zip** into **c:\str**, as shown in the picture below.



(*) please comply with directory, as it allows faster evaluation of your lab works.

RUN VISUAL STUDIO 19

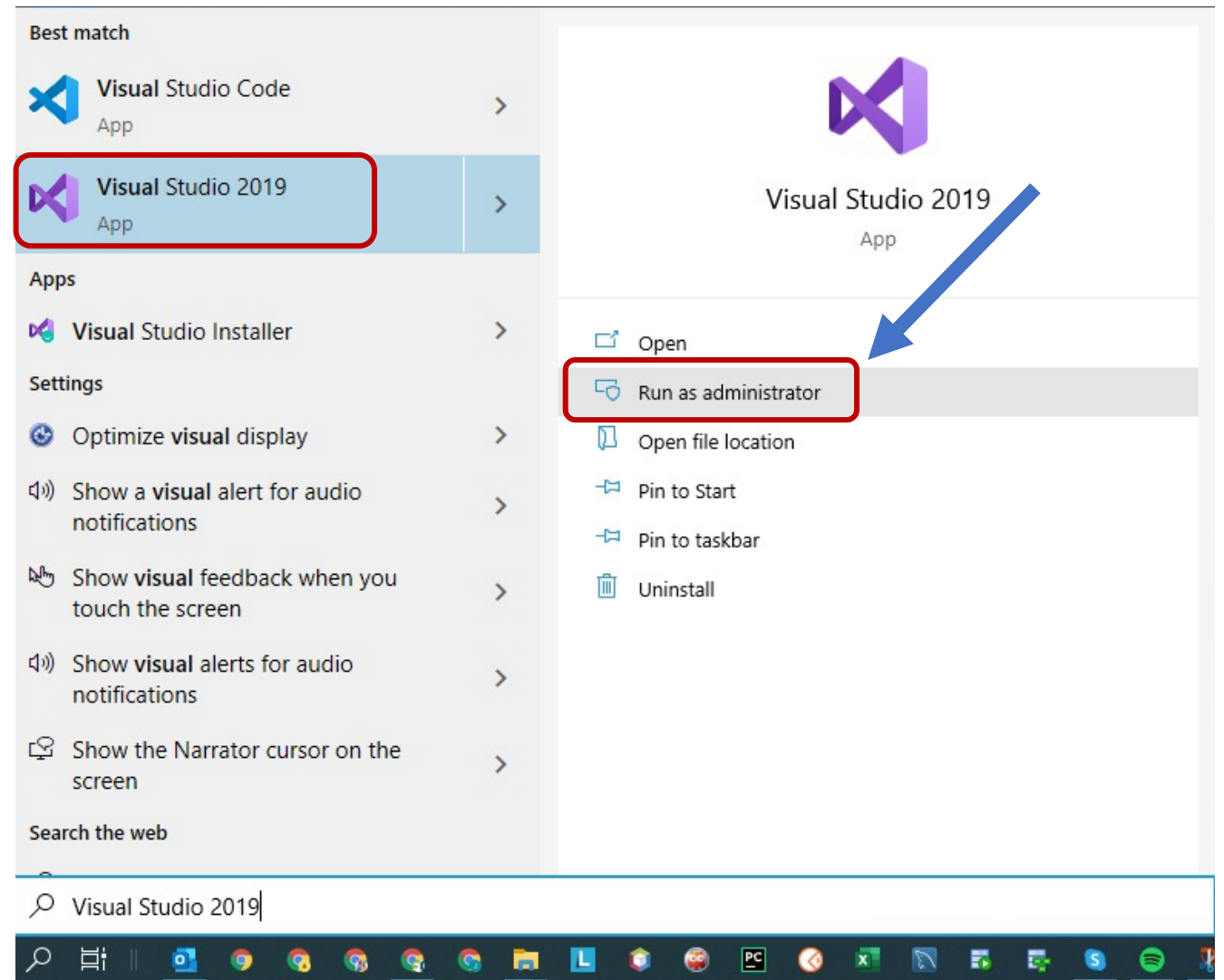


Run VS19

- Find Visual Studio 2019 in your startup menu
- Right click with the mouse
- Select “Run as administrator” (*)

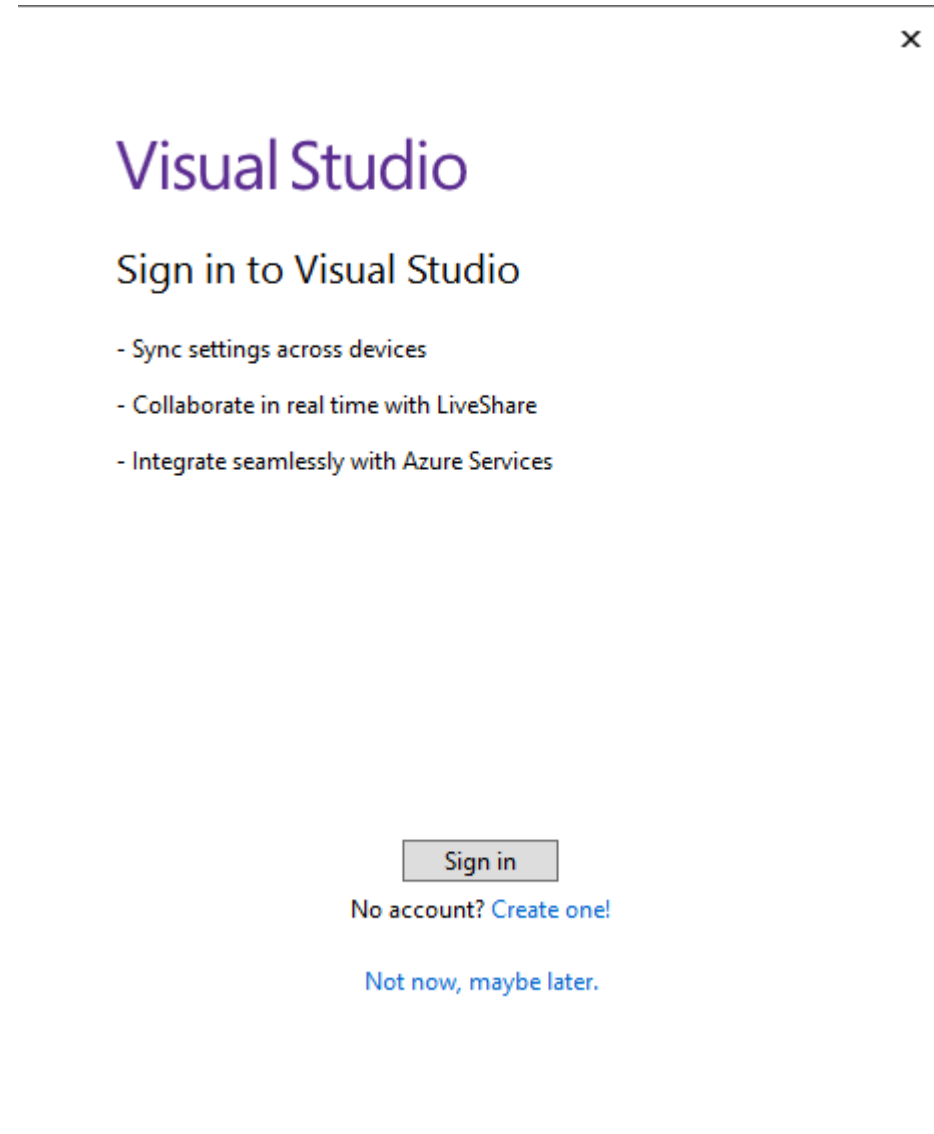
(*) This ensures that our project will obtain more privileges from the Operating System (Windows), ensuring reliable execution in a real time context.

This is important!

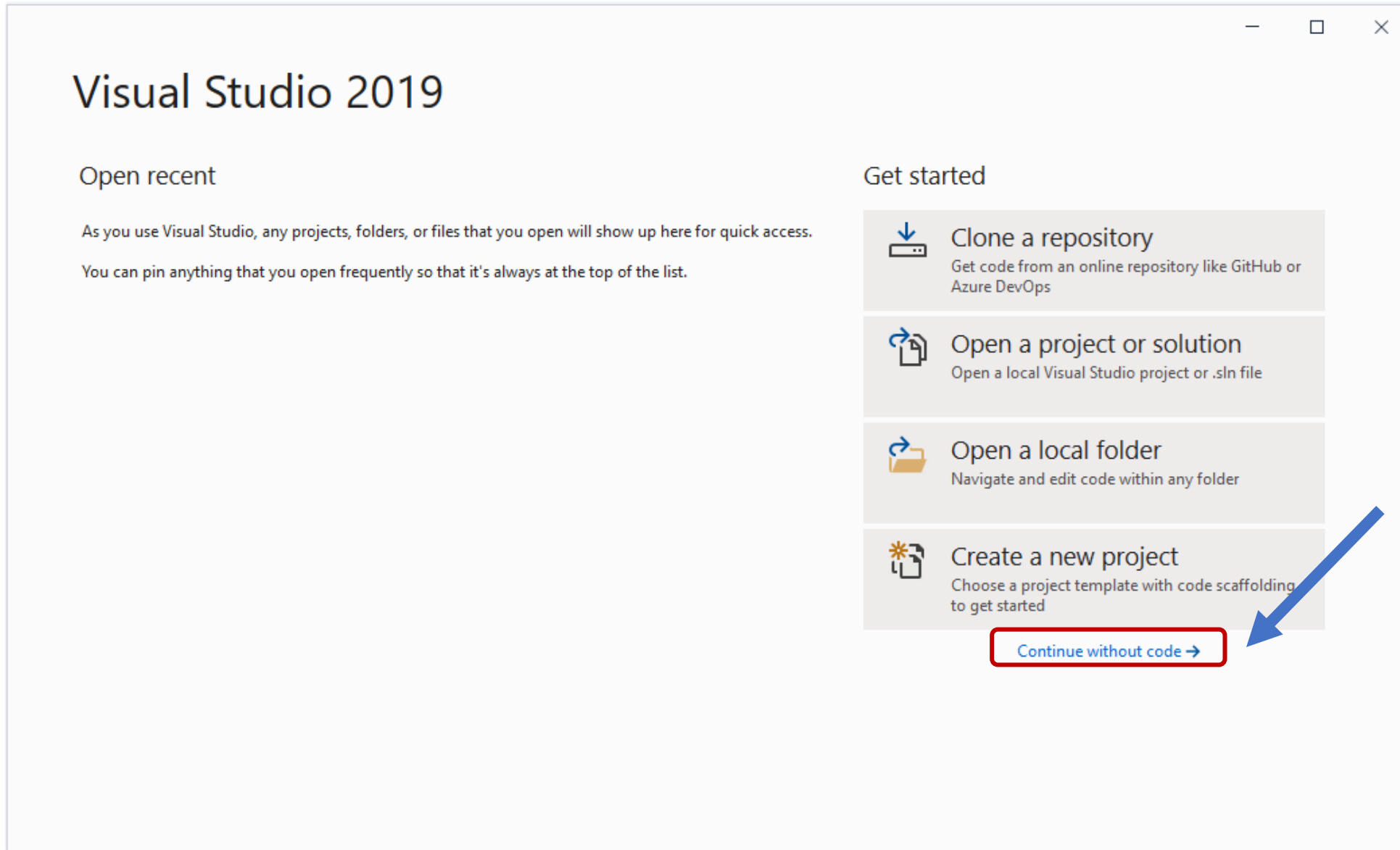


□ Signing in

- “Sign in” with your faculty credentials (student login and password) or select “not now”.
- For the credentials, note that it is necessary to provide them as student_id@fct.unl.pt **(without “campus”)**.



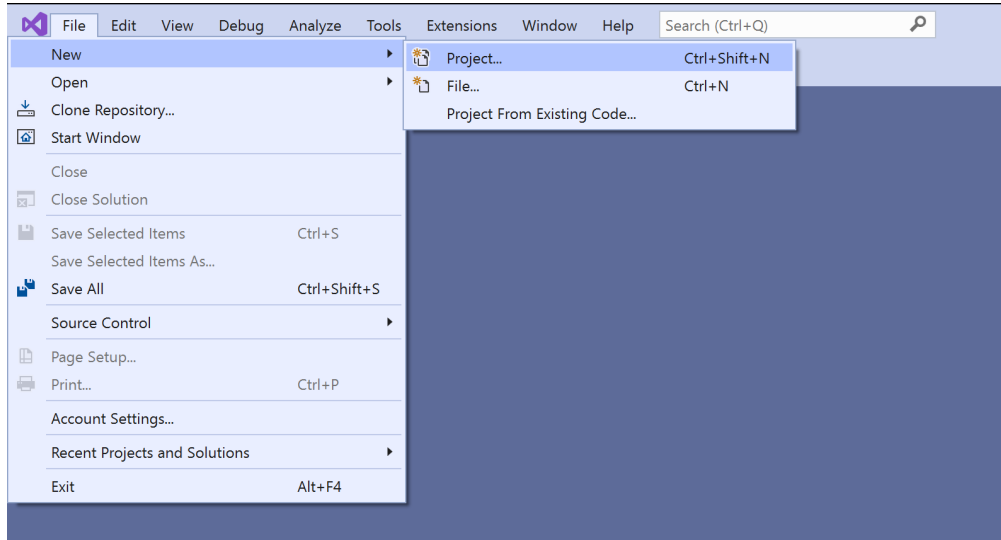
CREATE LABWORK1 PROJECT



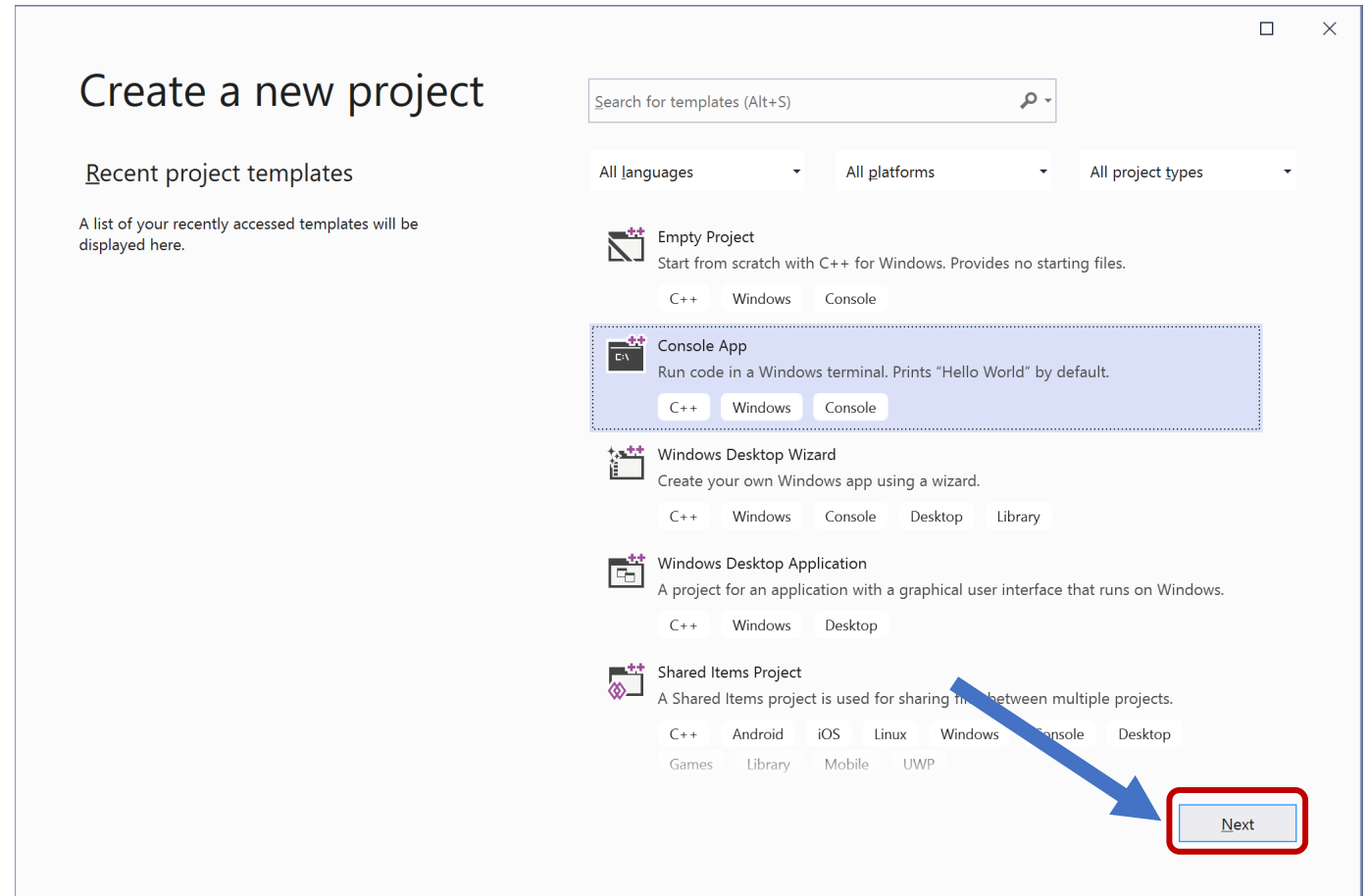
CREATE LABWORK1 PROJECT



Click File > New > Project



Select **Console App** project



CREATE LABWORK1 PROJECT



☐ Project name: **labwork1**

☐ Location: **c:\str**

☐ Click on the checkbox.

Configure your new project

Console App C++ Windows Console

Project name

labwork1

Location

C:\str\

Solution name ⓘ

labwork1

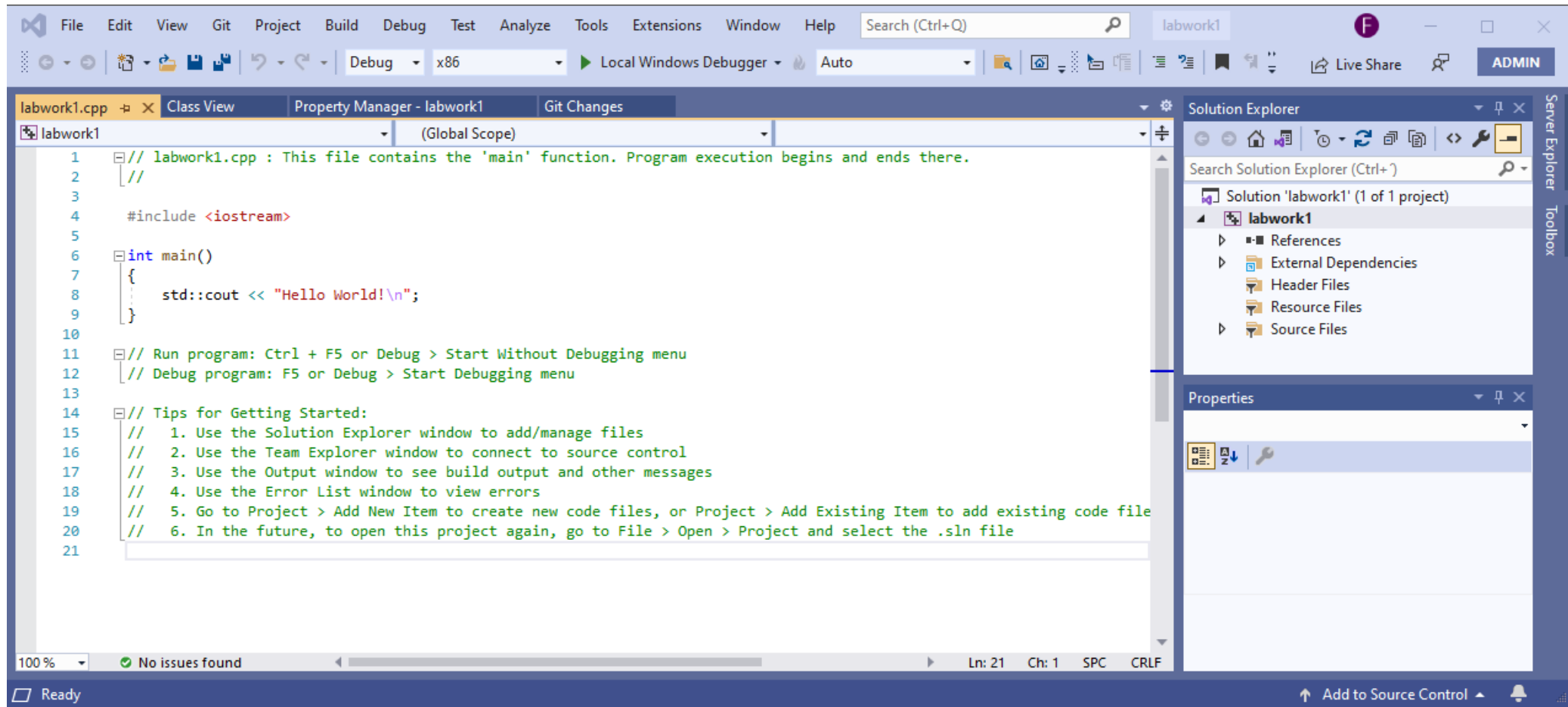
☒ Place solution and project in the same directory

Back Create

CREATE LABWORK1 PROJECT

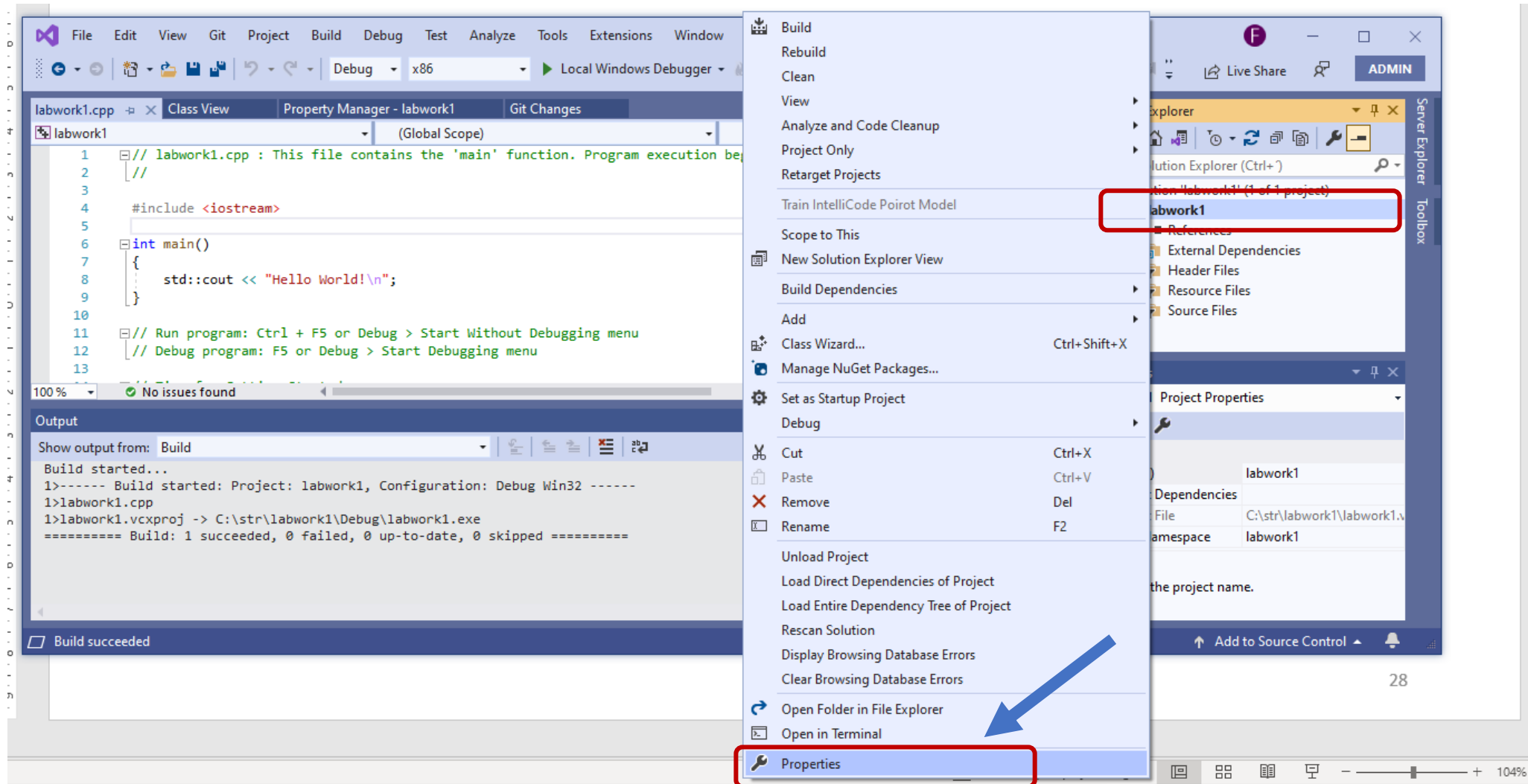


- ❑ Your console App project



CONFIGURE PROJECT PROPERTIES

❏ Right-click **labwork1** ➡ Select **Properties**



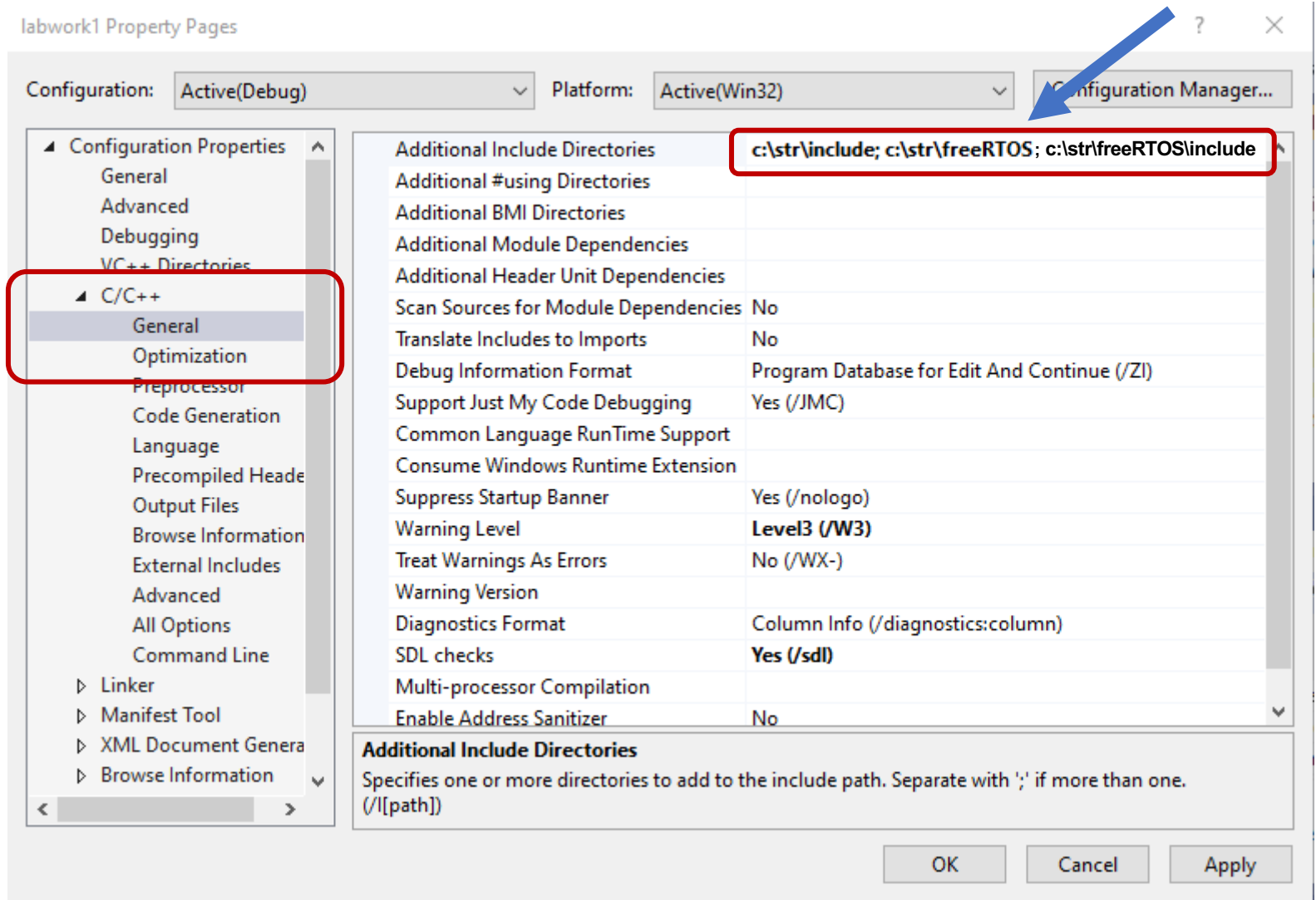
CONFIGURE PROJECT PROPERTIES



☐ Select C/C++ -> General

☐ Copy the include directories:

c:\str\include; c:\str\freeRTOS;
c:\str\FreeRTOS\include

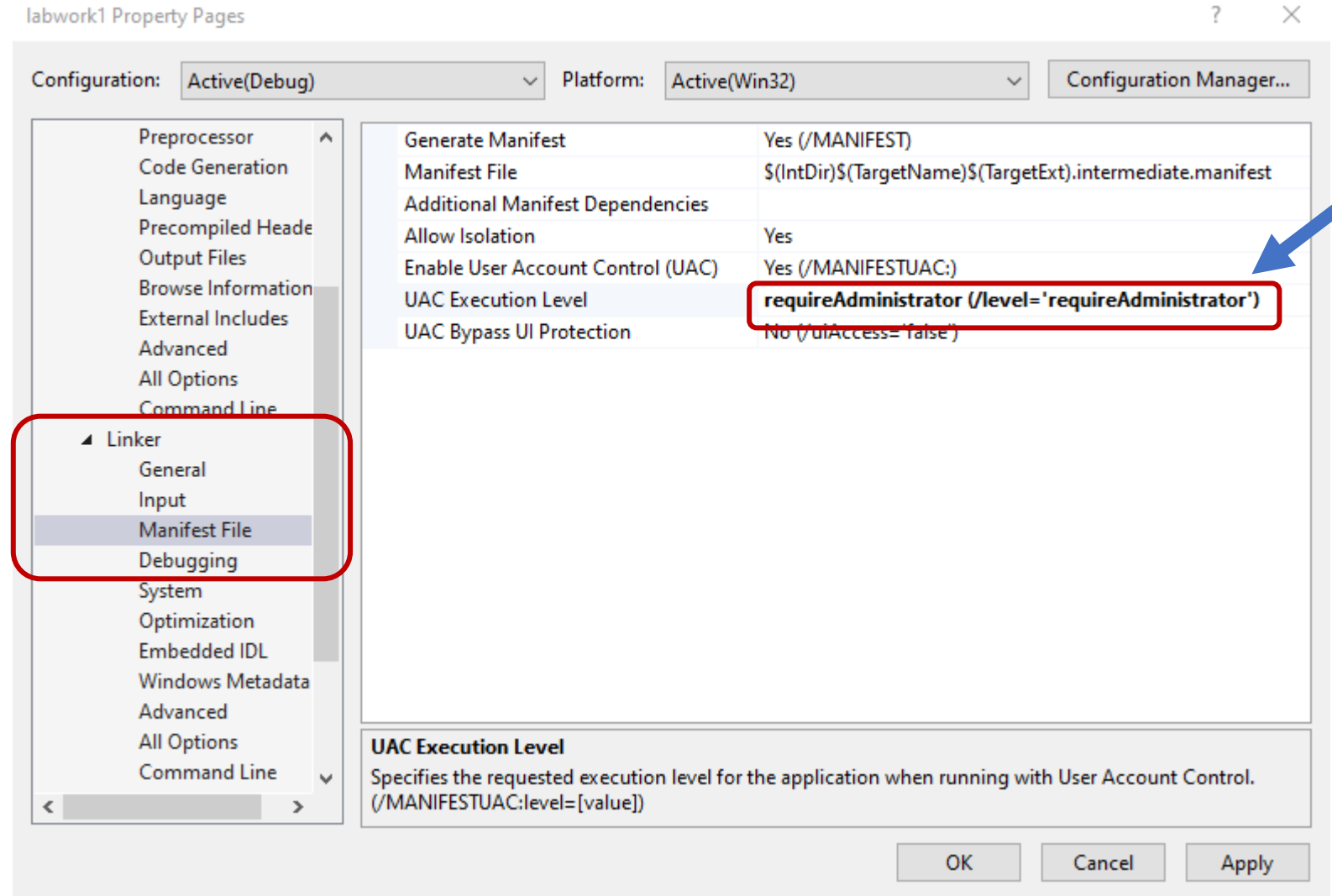


CONFIGURE PROJECT PROPERTIES



- ❑ Select Linker -> Manifest File
- ❑ Change UAC Execution level to “requireAdministrator” (*)

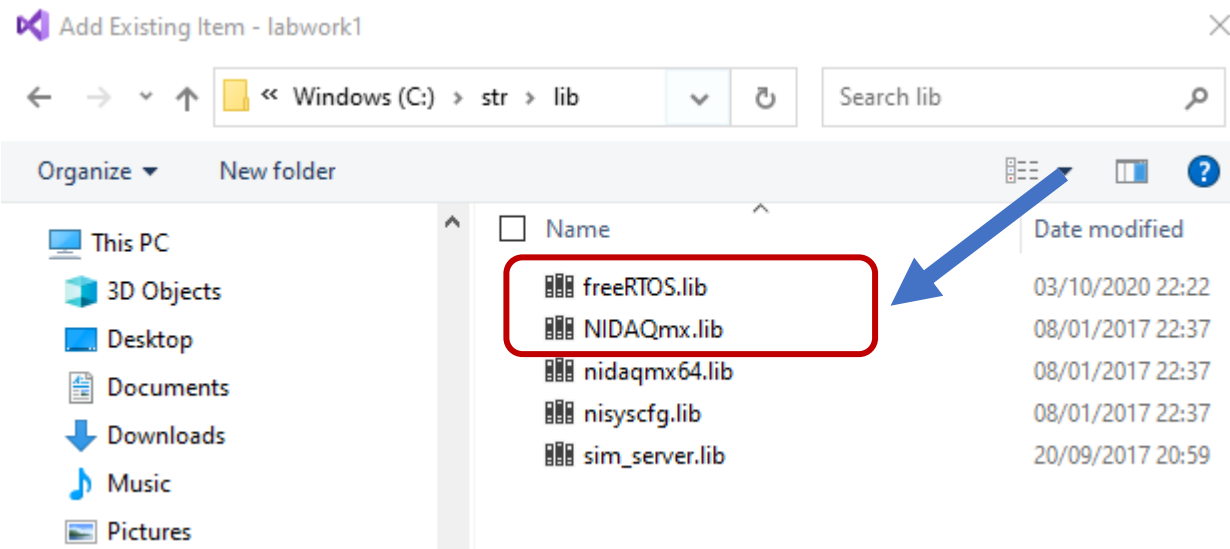
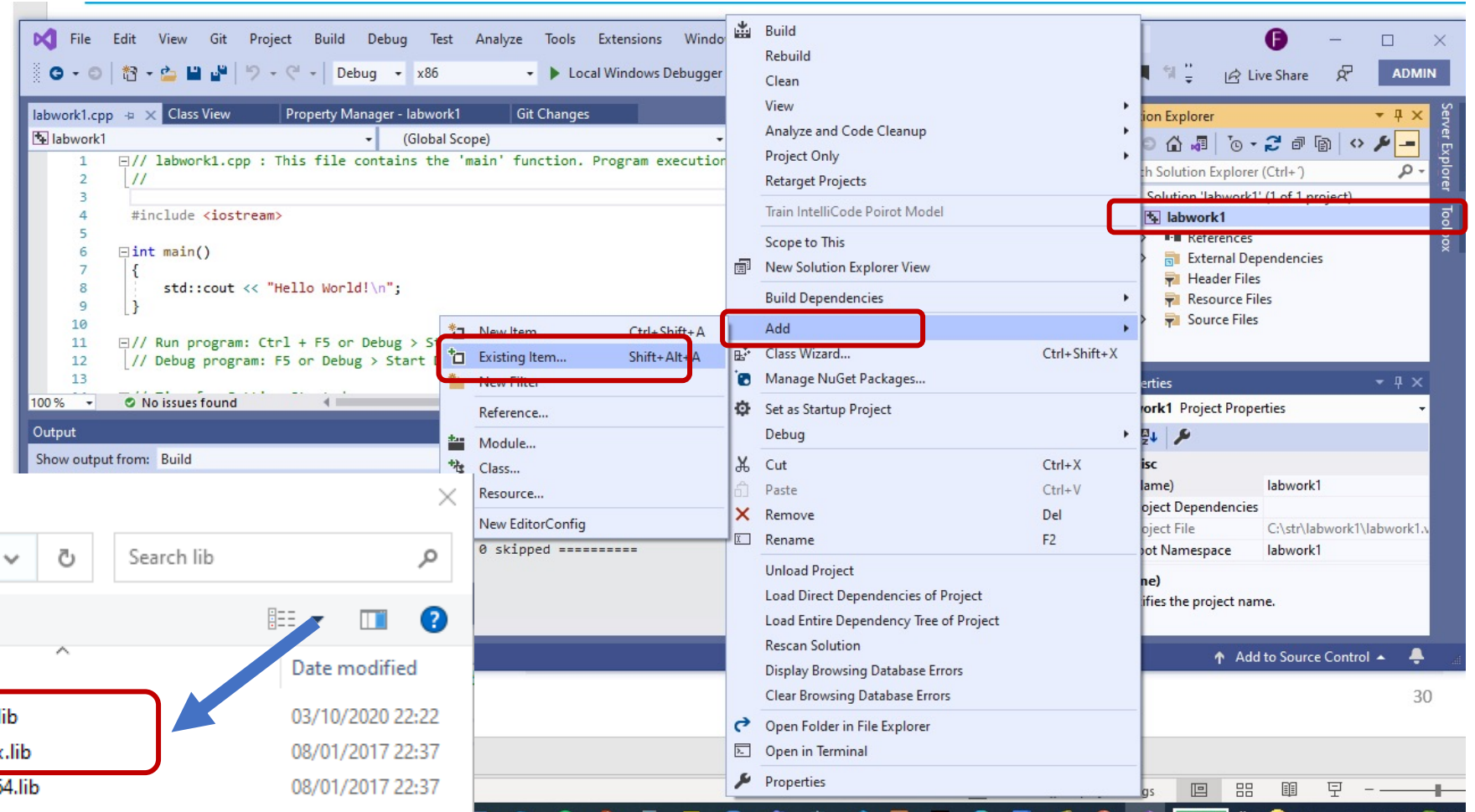
(*) As mentioned before, your application will receive more privileges from Windows O.S., necessary for adequate real-time behavior.



CONFIGURE PROJECT PROPERTIES



- ❑ Right Click in the project and select: Add > Existing item.
- ❑ In folder c:\str\lib, select the files below.
- ❑ Click Add button.

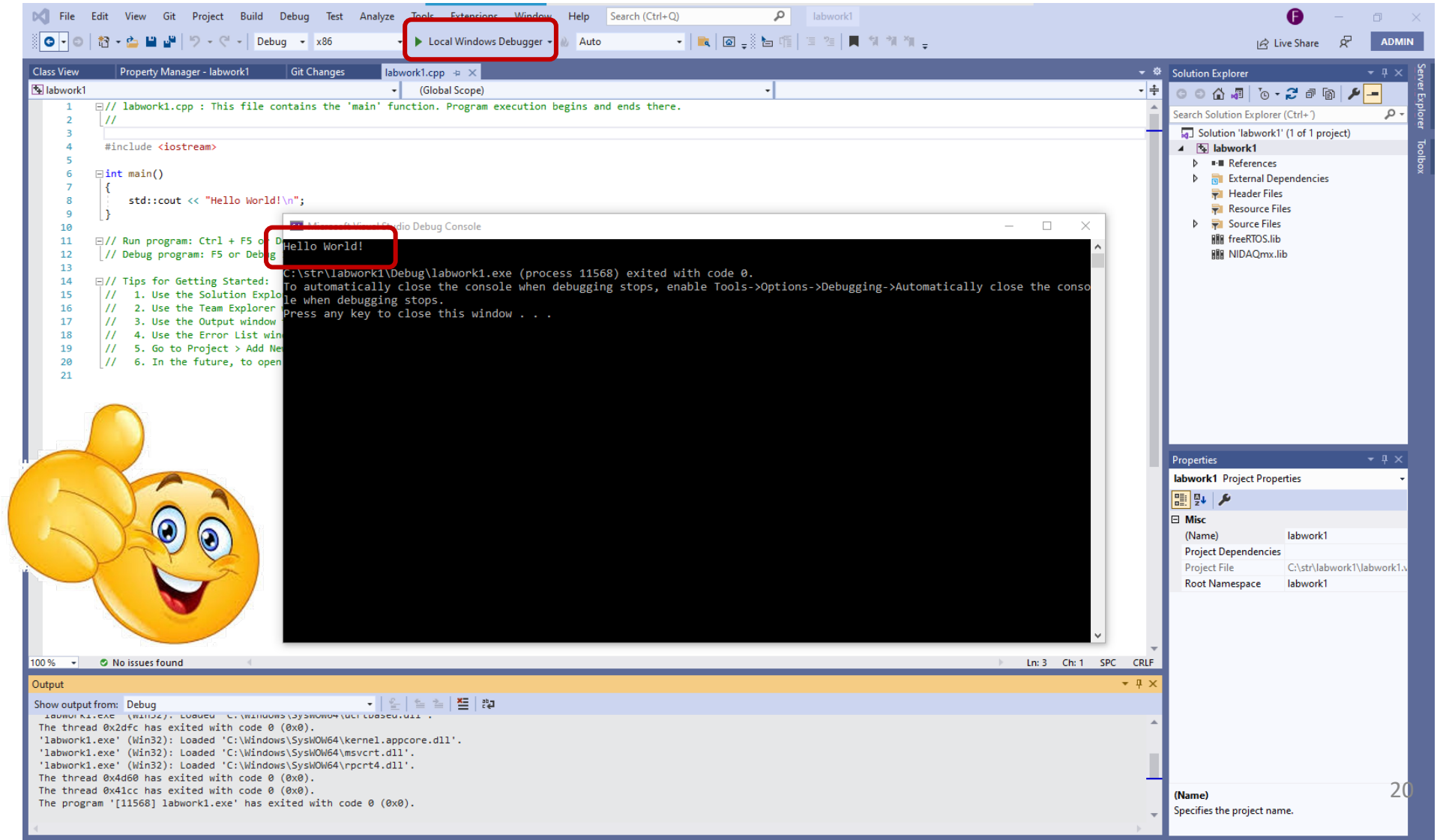


RUN PROJECT



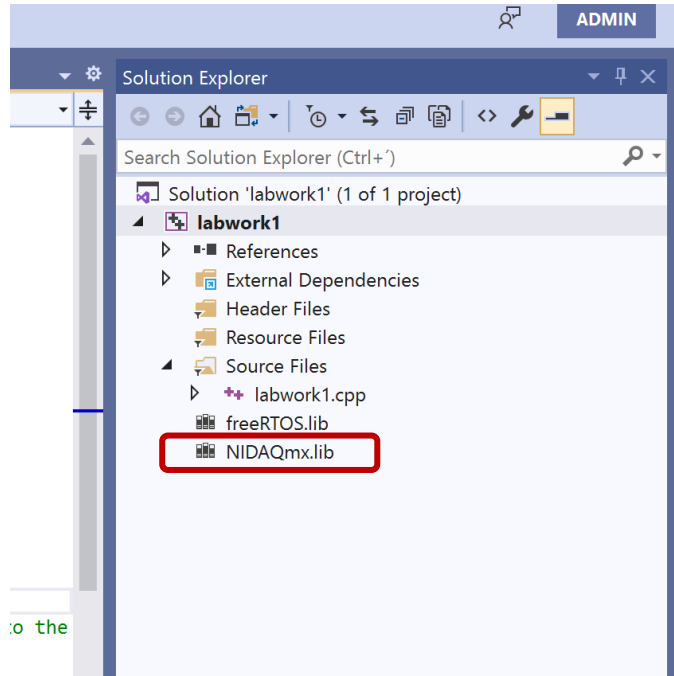
❑ Execute project (Click “Local Windows Debugger” or press F5).

❑ Check if it runs



PART 2

Interaction with Physical System

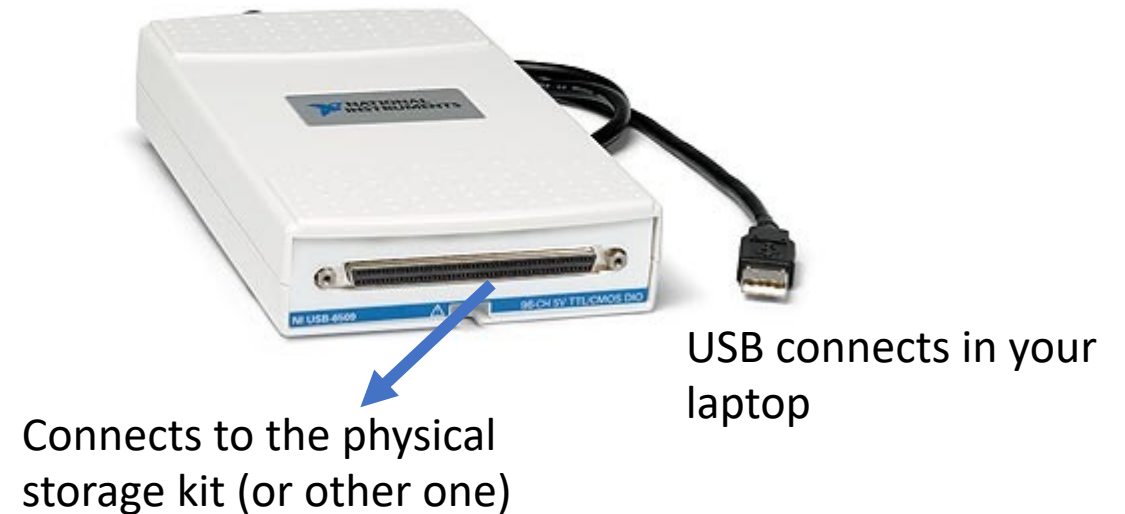


NIDAQmx.lib, which is the library for letting your application interact with the storage kit, through an acquisition board...

The NI USB-6509 DAQ Board

□ A few features

- Data acquisition (DAQ).
- The NI USB-6509 features 96 TTL/CMOS-compatible digital I/O lines,
- Watchdog timer, digital input filtering and change detection...
- Output programmable power-up states.
- ...



DIGITAL ACQUISITION BOARD



The NI USB-6509 DAQ Board



Figure 6. Pin Assignments for the NI USB-6509 using the SH100-100-F

Pin assignment:

P2.7	1	51	P8.7
P5.7	2	52	P11.7
P2.6	3	53	P8.6
P5.6	4	54	P11.6
P2.5	5	55	P8.5
P5.5	6	56	P11.5
P2.4	7	57	P8.4
P5.4	8	58	P11.4
P2.3	9	59	P8.3
P5.3	10	60	P11.3
P2.2	11	61	P8.2
P5.2	12	62	P11.2
P2.1	13	63	P8.1
P5.1	14	64	P11.1
P2.0	15	65	P8.0
P5.0	16	66	P11.0
P4.7	17	67	P7.7
P1.7	18	68	P10.7
P1.6	19	69	P7.6
P4.6	20	70	P10.6
P1.5	21	71	P7.5
P4.5	22	72	P10.5
P1.4	23	73	P7.4
P4.4	24	74	P10.4
P1.3	25	75	P7.3
P4.3	26	76	P10.3
P1.2	27	77	P7.2
P4.2	28	78	P10.2
P1.1	29	79	P7.1
P4.1	30	80	P10.1
P1.0	31	81	P7.0
P4.0	32	82	P10.0
P0.7	33	83	P6.7
P3.7	34	84	P6.7
P0.6	35	85	P6.6
P3.6	36	86	P6.6
P0.5	37	87	P6.5
P3.5	38	88	P6.5
P0.4	39	89	P6.4
P3.4	40	90	P6.4
P0.3	41	91	P6.3
P3.3	42	92	P6.3
P0.2	43	93	P6.2
P3.2	44	94	P6.2
P0.1	45	95	P6.1
P3.1	46	96	P6.1
P0.0	47	97	P6.0
P3.0	48	98	P6.0
+5 V	49	99	+5 V
GND	50	100	GND

The NI USB-6509 DAQ Board

- ❑ Left side lies the 96 DIO, which can be set programmatically as input/output inside your C++ code.
- ❑ For Diagnostic and testing, the NI MAX App can be used.

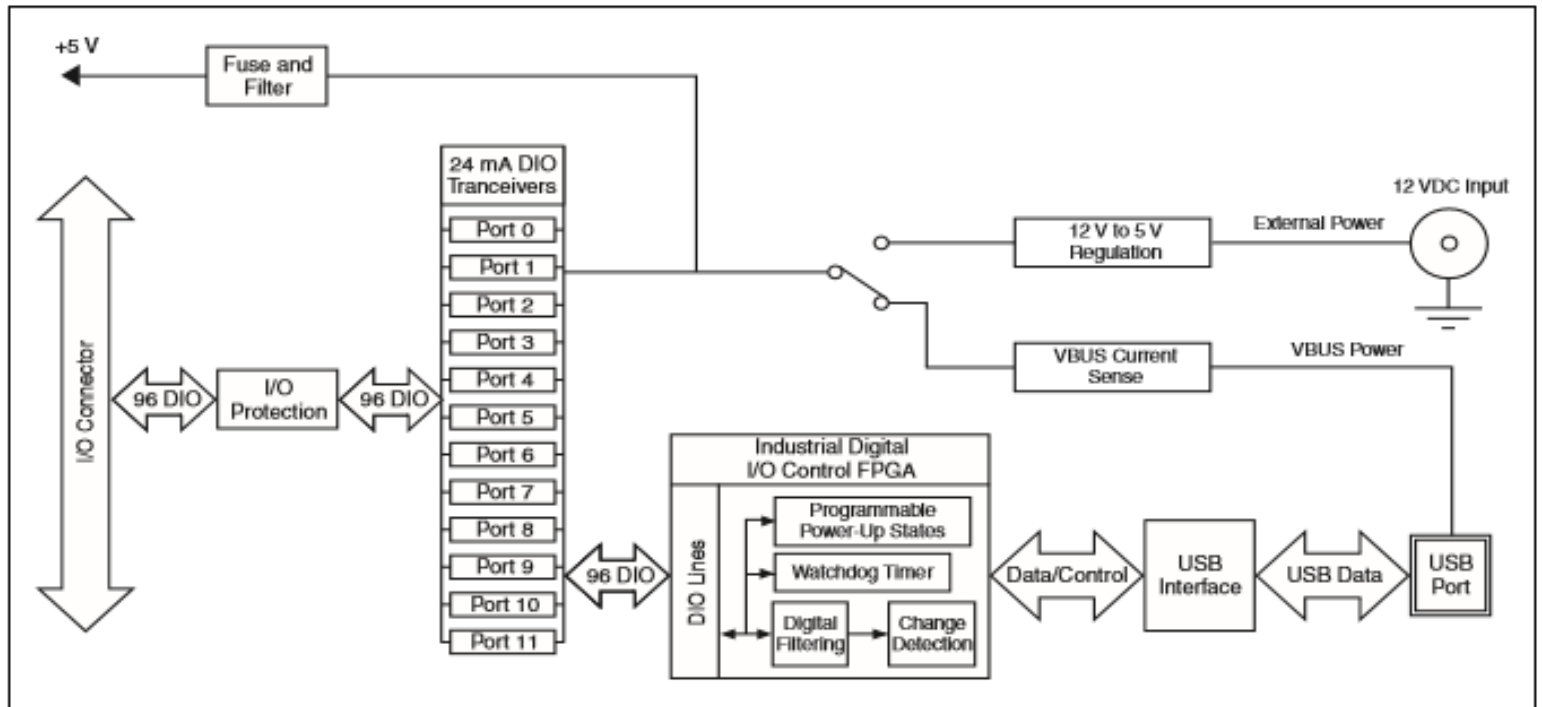
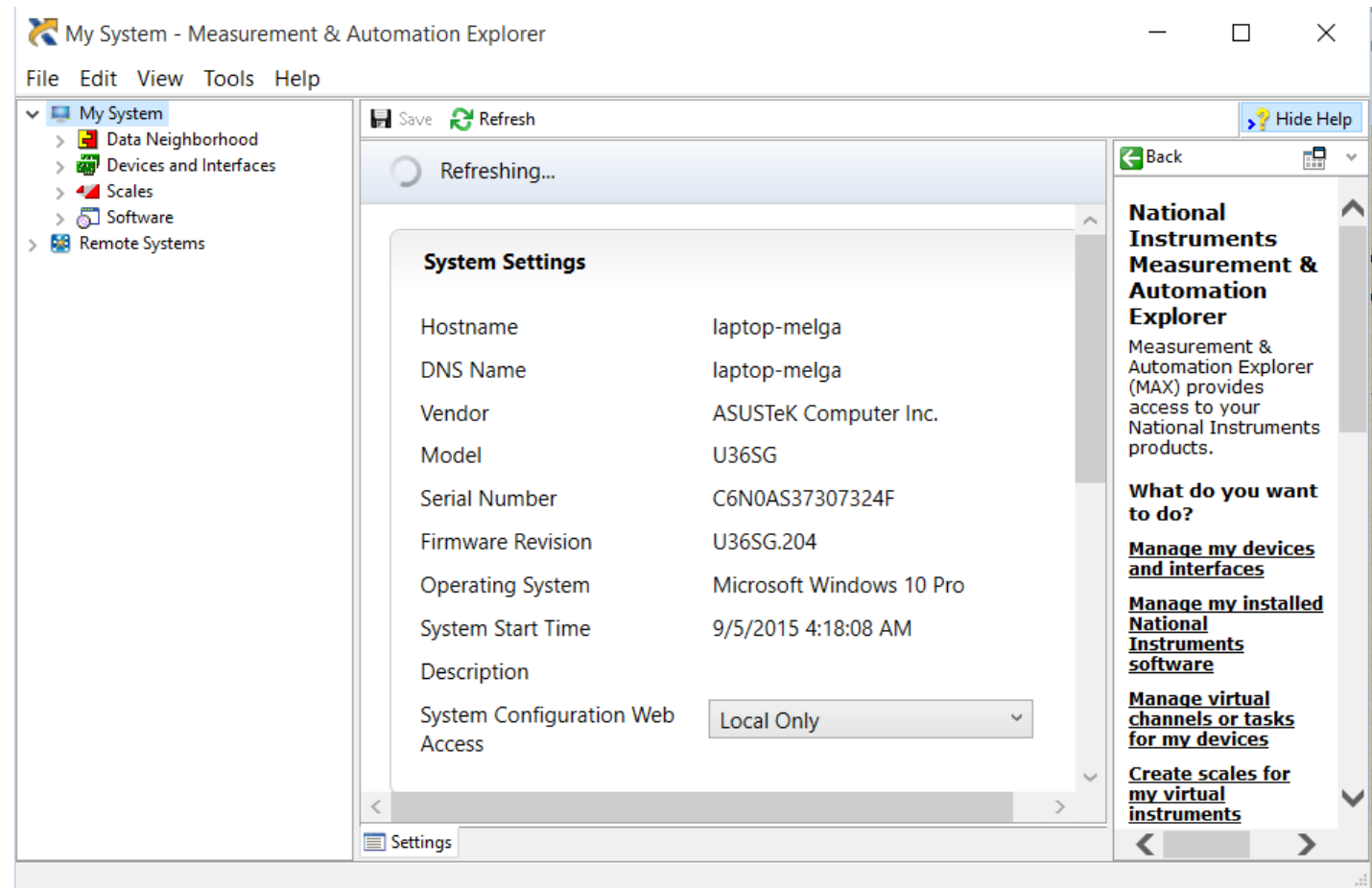


Figure 4. NI USB-6509 Block Diagram

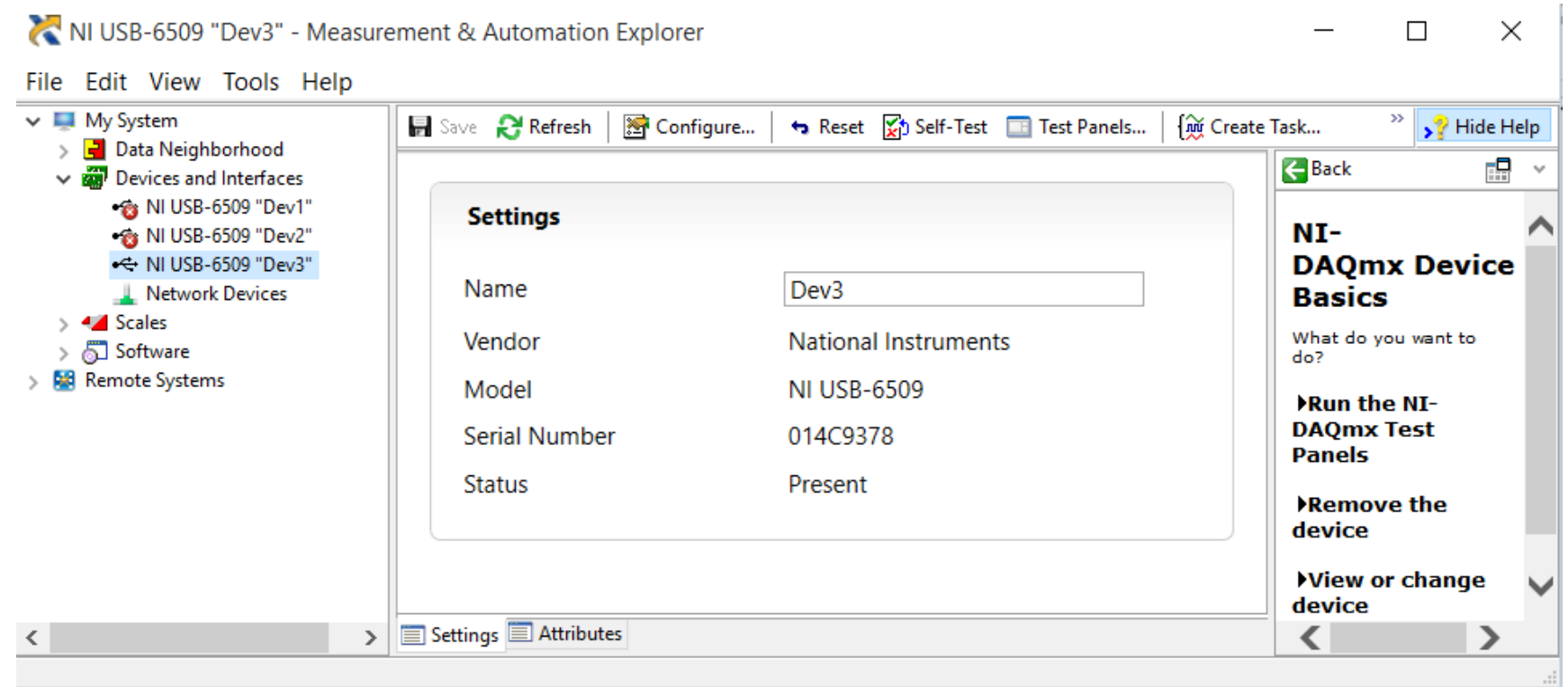
The NI USB-6509 DAQ Board

- ❑ Device configuration application



The NI USB-6509 DAQ Board

- Device selection

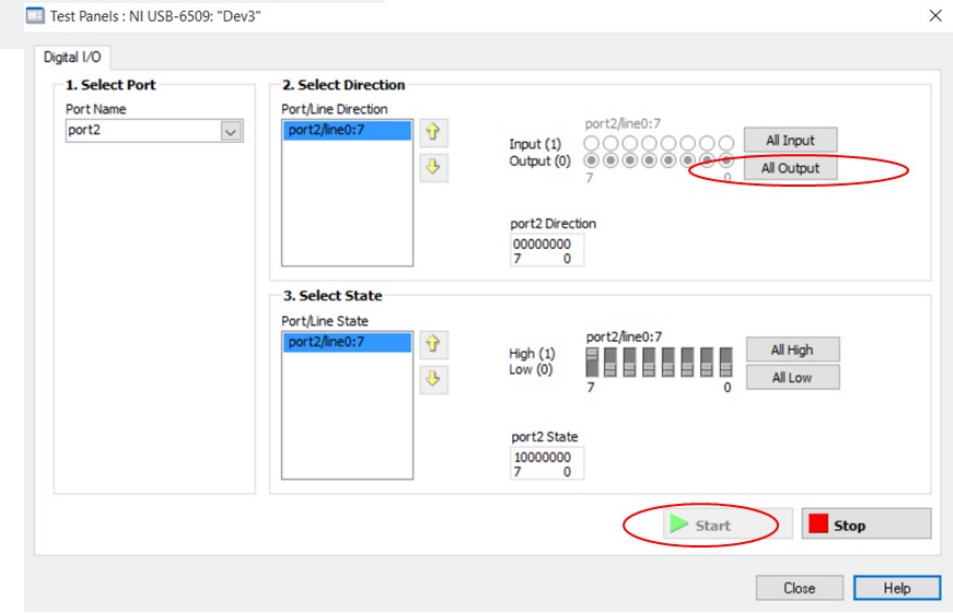
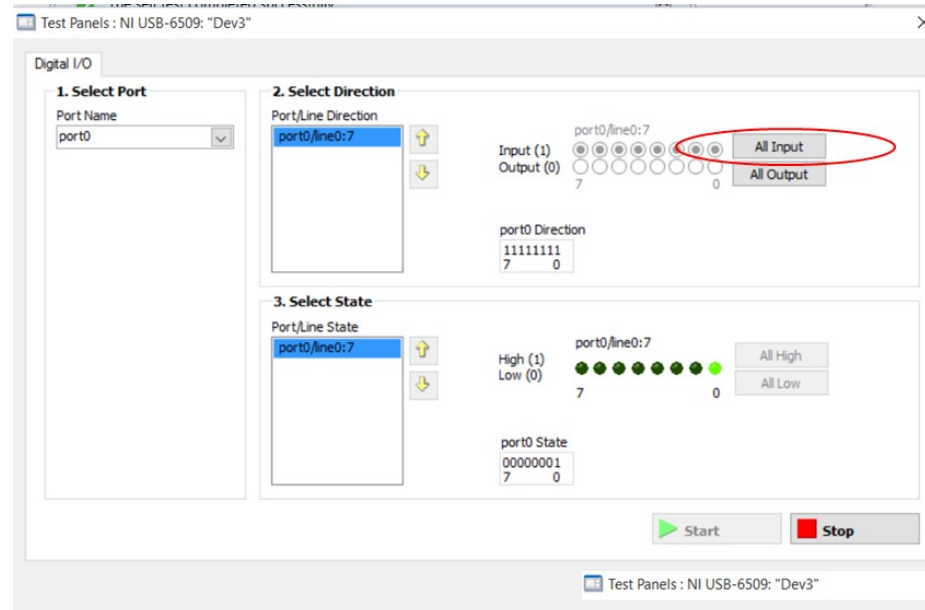


DIGITAL ACQUISITION BOARD



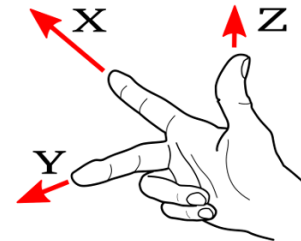
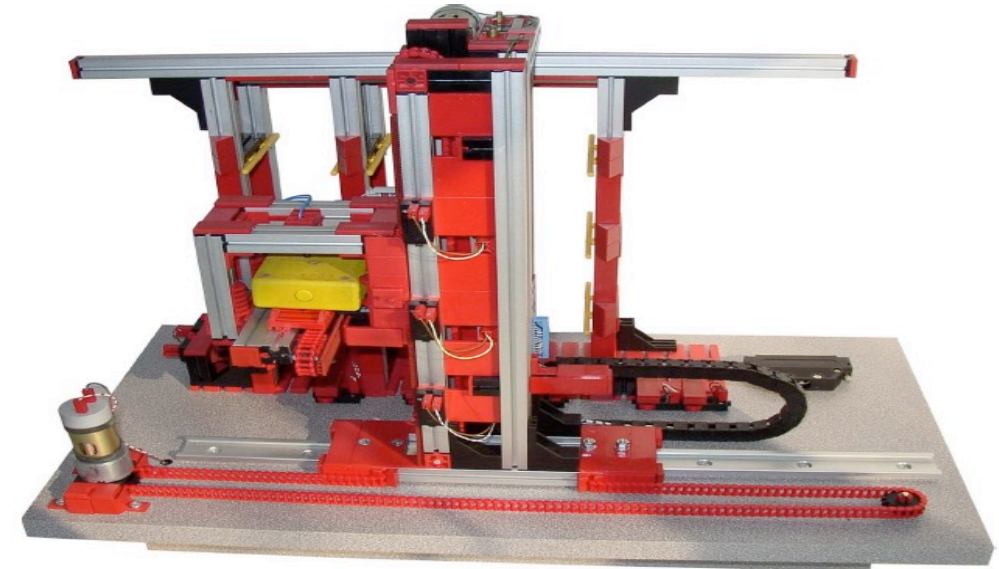
The NI USB-6509 DAQ Board

- ❑ Test panels: Port 0 and 1: all inputs
- ❑ set port 0 and 1 bits as inputs
- ❑ set port 2 bits as outputs



The physical storage kit

- ❑ Here is a picture of the storage kit.
- ❑ We can move the cage with a pallet on the xx , yy , zz axis with directions according to the right-hand rule.
- ❑ Pallets put or taken from the cells.

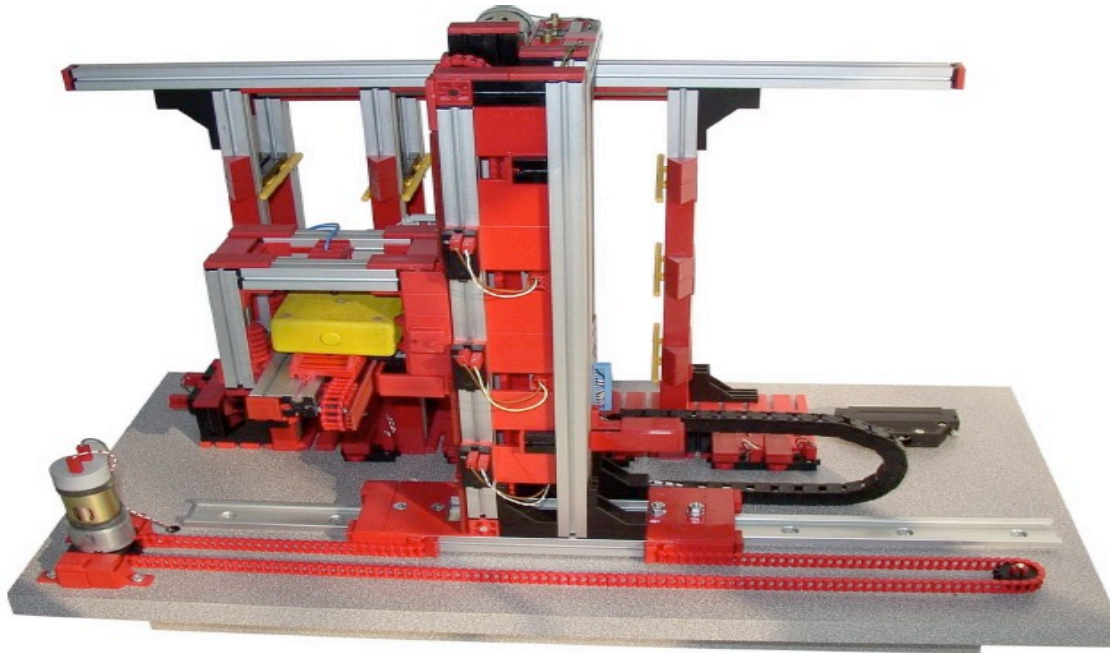


<https://www.ias.informatik.tu-darmstadt.de/Teaching/RobotLearningLecture2>

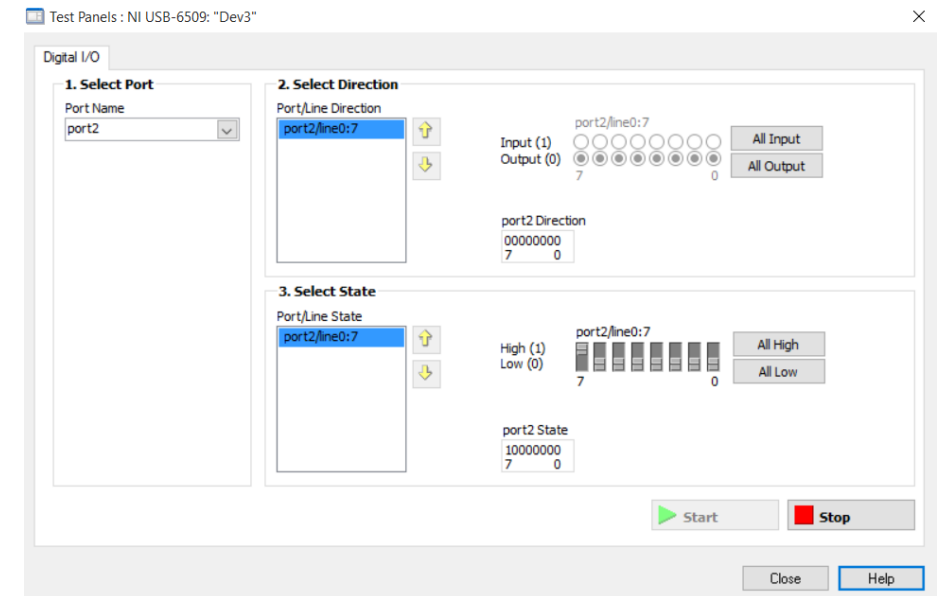
THE PHYSICAL STORAGE KIT



Activating a bit in port 2

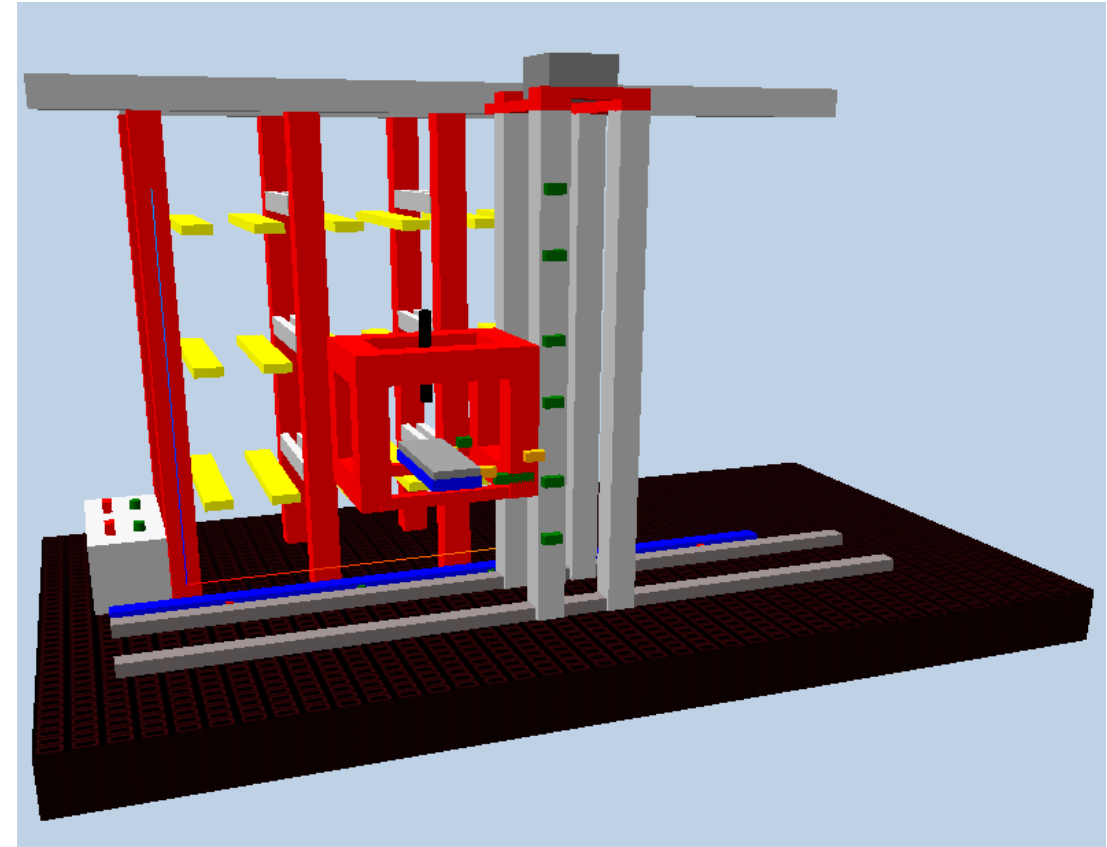


- ❑ If we set $P2.7 = 1$, the storage starts moving to the right.
- ❑ We are going to learn how to do it in your C++ program



Using the Simulator

- ☐ There is a simulator to replace the physical storage kit to overcome remote classes context (sometimes, even in classroom, due to the number of students)
- ☐ All functionality and coding errors can be solved using the simulator.
- ☐ Switching from kit to simulator could be done without any changes in your C++ code.



CONFIGURING LABWORK1 PROJECT PROPERTIES



More Project Configuration

- It tells the compiler to look for *.h
include files in its **own project folder**

LabWork1 Property Pages

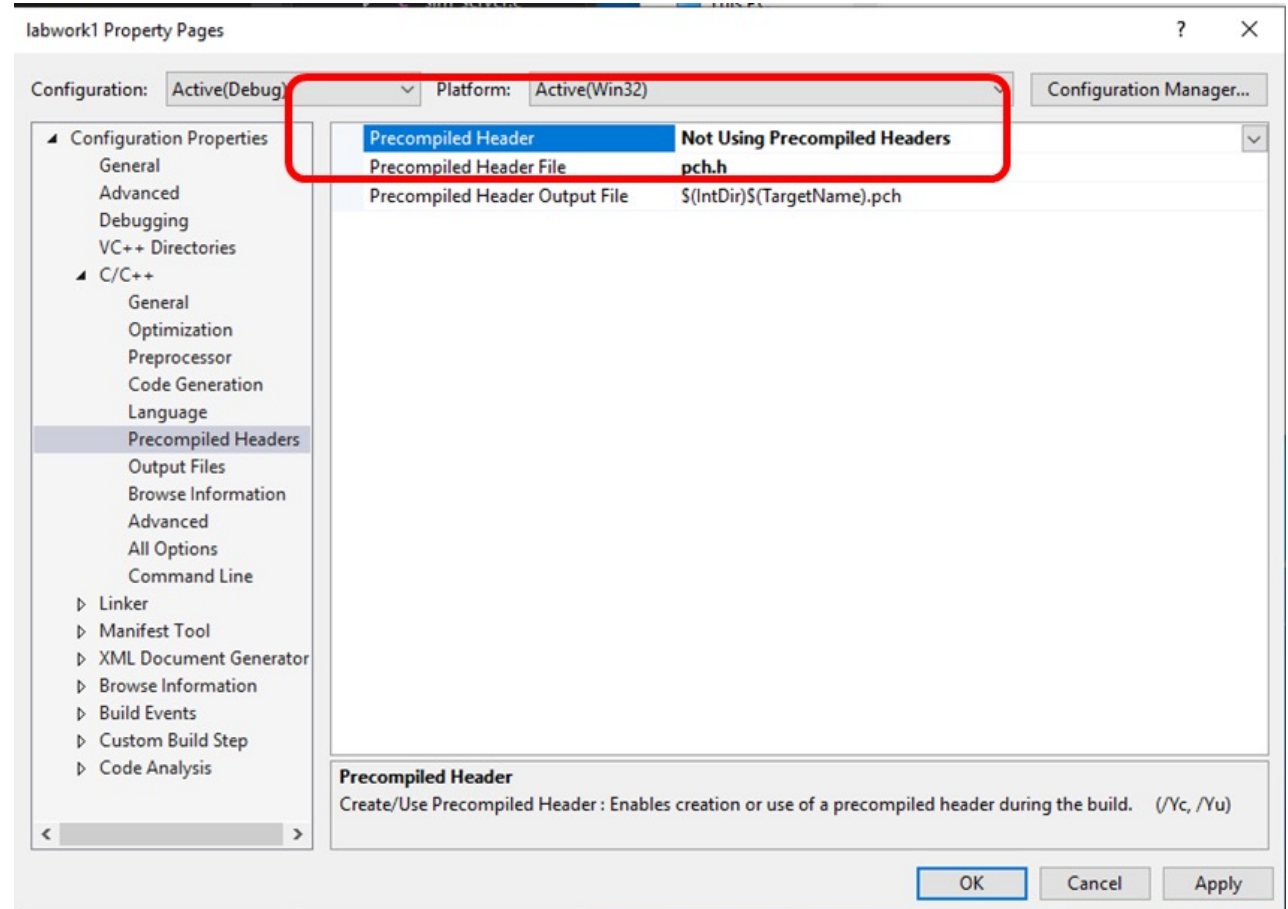
Configuration:	Active(Debug)	Platform:	Active(Win32)
Configuration Properties			
General			
Debugging			
VC++ Directories			
C/C++			
General			
Optimization			
Preprocessor			
Code Generation			
Language			
Precompiled Headers			
Output Files			
Browse Information			
Additional Include Directories			;c:\str2018\include;c:\s
Additional #using Directories			
Debug Information Format			Program Database for Ec
Support Just My Code Debugging			Yes (/JMC)
Common Language RunTime Support			
Consume Windows Runtime Extension			
Suppress Startup Banner			Yes (/nologo)
Warning Level			Level3 (/W3)
Treat Warnings As Errors			No (/WX-)
Warning Version			
Diagnostics Format			Classic (/diagnostics:clas
SDL checks			Yes (/sdl)

CONFIGURING LABWORK1 PROJECT PROPERTIES



More Project Configuration

- Change pre-compiled headers option
- Select “Not Using Precompiled Headers”
- Using it would be important in big projects (with many modules)



CONFIGURING LABWORK1 PROJECT PROPERTIES



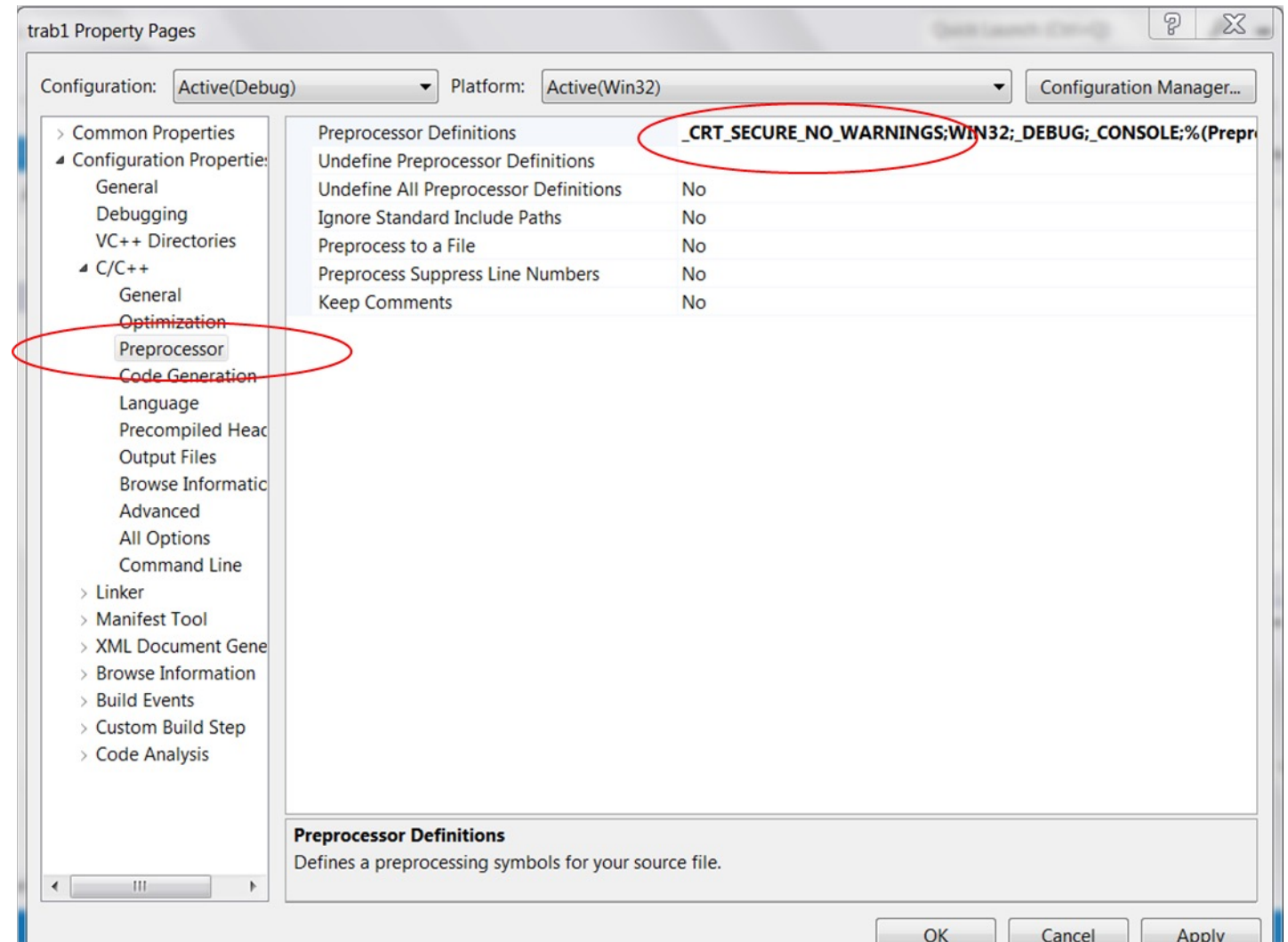
More Project Configuration

- Configure (add)

“_CRT_SECURE_NO_WARNINGS”
(in Project properties)

Copy past:

_CRT_SECURE_NO_WARNINGS;

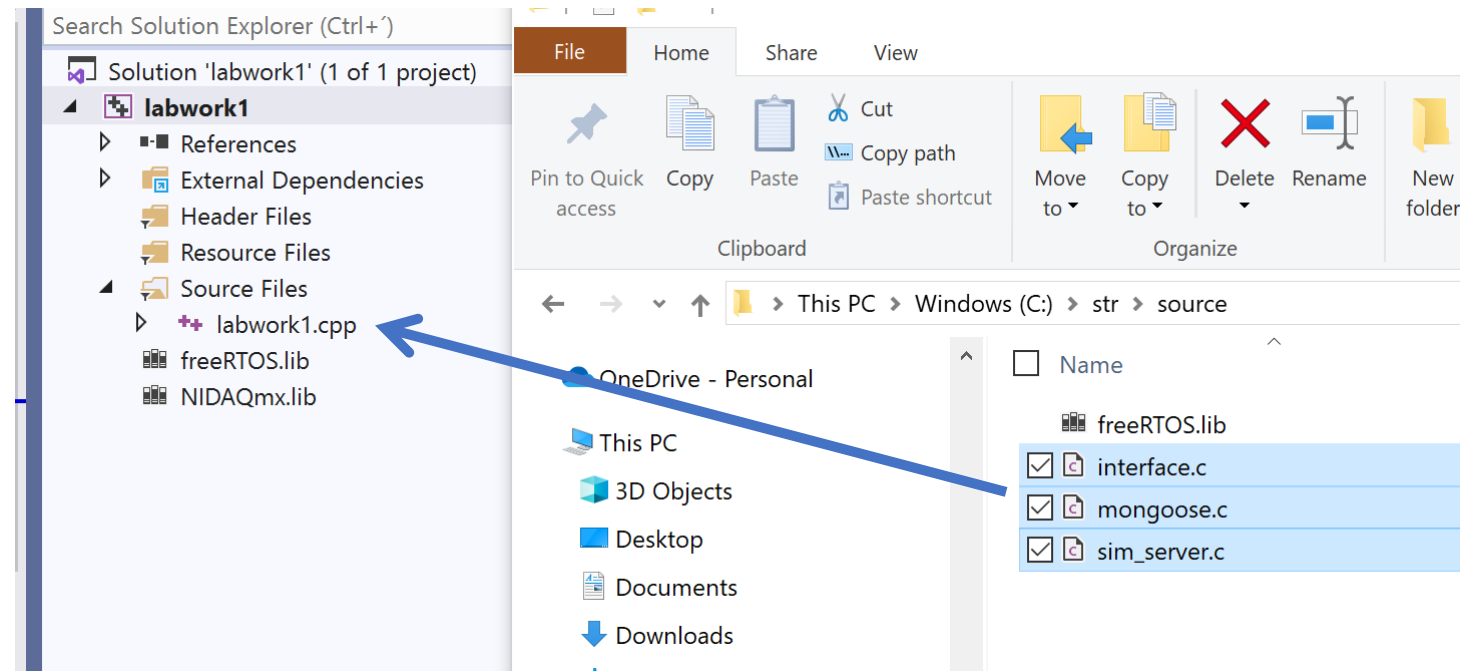


LABWORK1 PROJECT



□ Add source code to your project

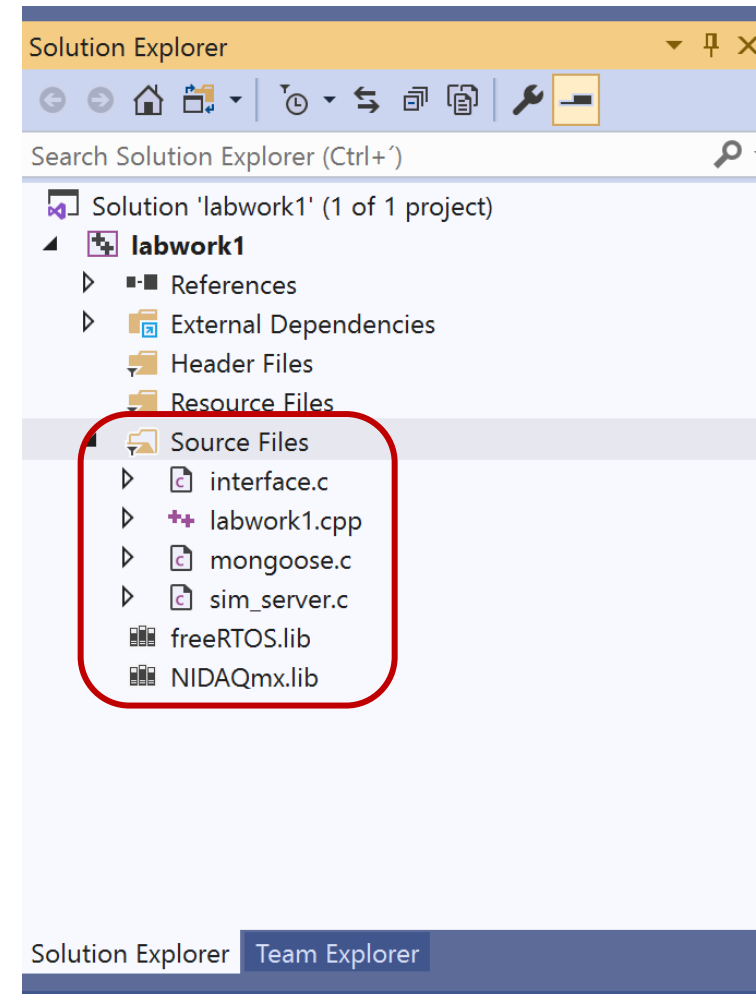
- Drag and drop as illustrated below.
- Alternatively, right click on source Files > Add > Existing Items



LABWORK1 PROJECT



- Aspect of your solution explorer in VS19
 - **sim_server.c** and **mongoose.c** handles the interaction with the kit simulator
 - **NIDAQmx.lib** allows your code to connect with the DAQ board.
 - **interface.c** allows switching between simulator and physical kit.
 - **FreeRTOS.lib** holds the realTime functionality for your code (next class)



LABWORK1 PROJECT: Example #1



□ Try this example:

```
#include <iostream>
#include<conio.h>
#include<stdlib.h>
#include<windows.h> // for the Sleep function
```

```
extern "C" {
    #include <interface.h>
}
```

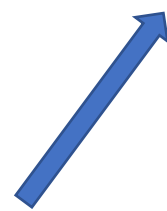
```
int main(int argc, char** argv) {
    createDigitalInput(0);
    createDigitalInput(1);
    createDigitalOutput(2);
```

```
    writeDigitalU8(2, 0x80); // start moving right
    Sleep(5000); //wait 5 seconds
    writeDigitalU8(2, 0);
    closeChannels();
}
```

Copy this code inside your source file (probably the labwork1.cpp)

It's important!

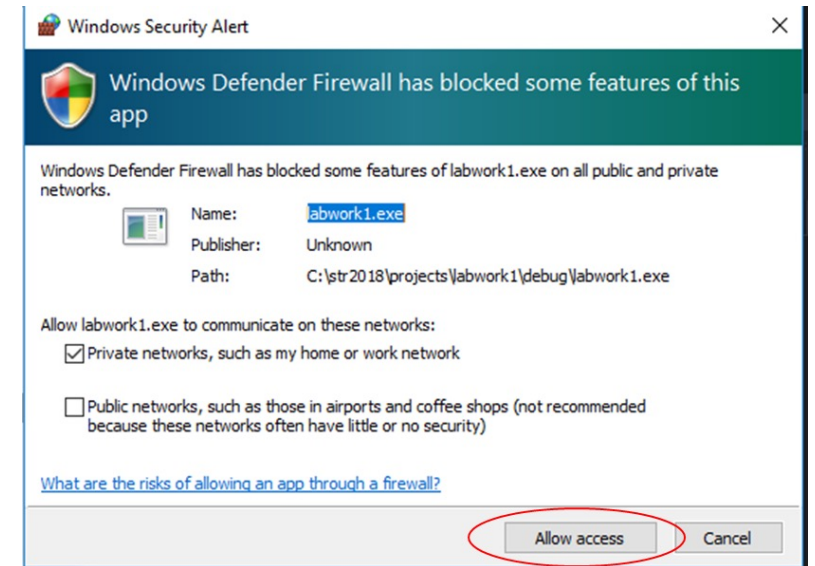
Allows using C and C++ together



□ Run

□ Allow the connection

□ It lets your program create a connection with the simulator



LABWORK1 PROJECT: the simulator



- ❑ In a browser (e.g. Chrome) try: http://localhost:8081/compact_storage.html
- ❑ Case page not found:
 - It means that simulator server is working, but page compact_storage.html is not found.
 - Probably your libraries for STR were copied into another folder, not the recommended one (c:\str\simulators)
 - To solve this issue, either:
 - Copy your libraries to the recommend folder or
 - Modify macro MG_HTTP_FOLDER inside interface.c to the value corresponding to the location of your own simulators folder
 - Case there is a port conflict, that is, port 8081 being already used, change to other value.

```
// interface.c
// J. Rosas

#include <stdlib.h>
#include <stdio.h>
#include <conio.h>
#include <signal.h>
#include <winsock.h>
#include <interface.h>

// #define _USE_REALKIT_
```

```
#define MG_HTTP_FOLDER "c:/str/simulators"
#define MG_HTTP_PORT "8081"
```

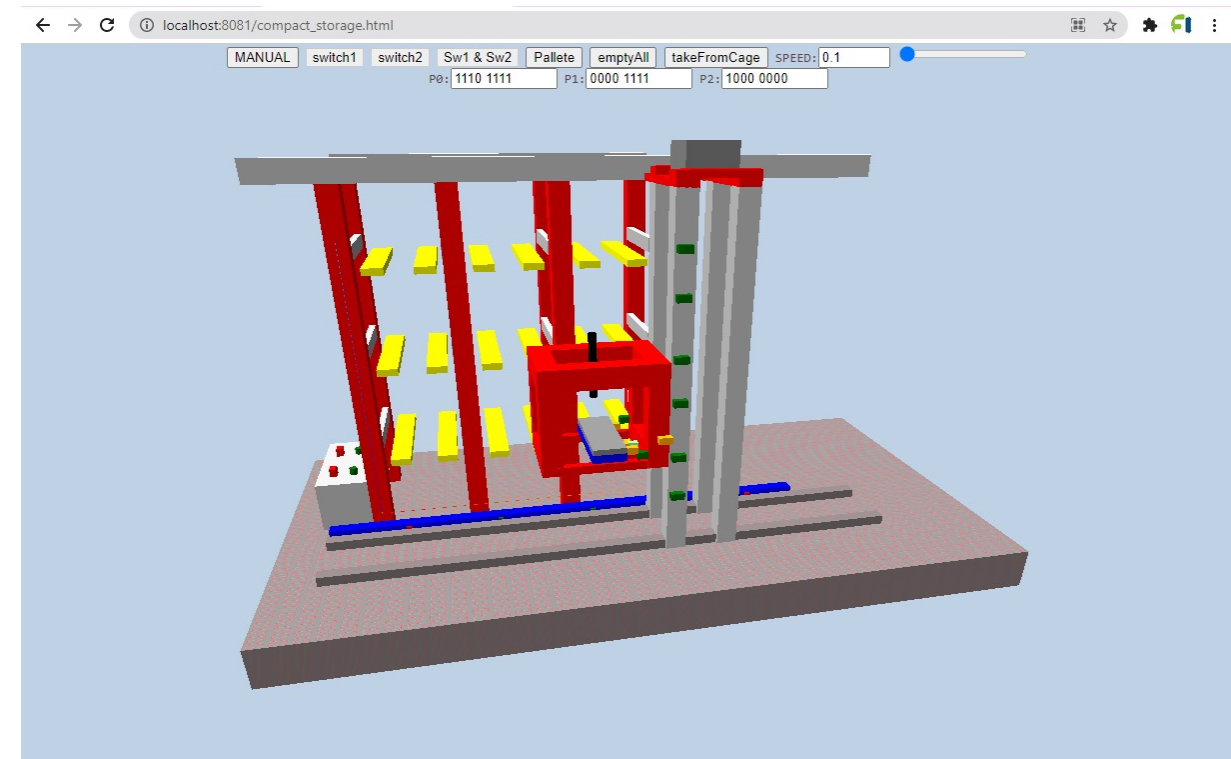
Not recommended!

LABWORK1 PROJECT: Run project



□ Run the labwork1 project

- Press F5 to run your project
- After it started running, copy the link http://localhost:8081/compact_storage.html into your browser bar address, to start the simulator

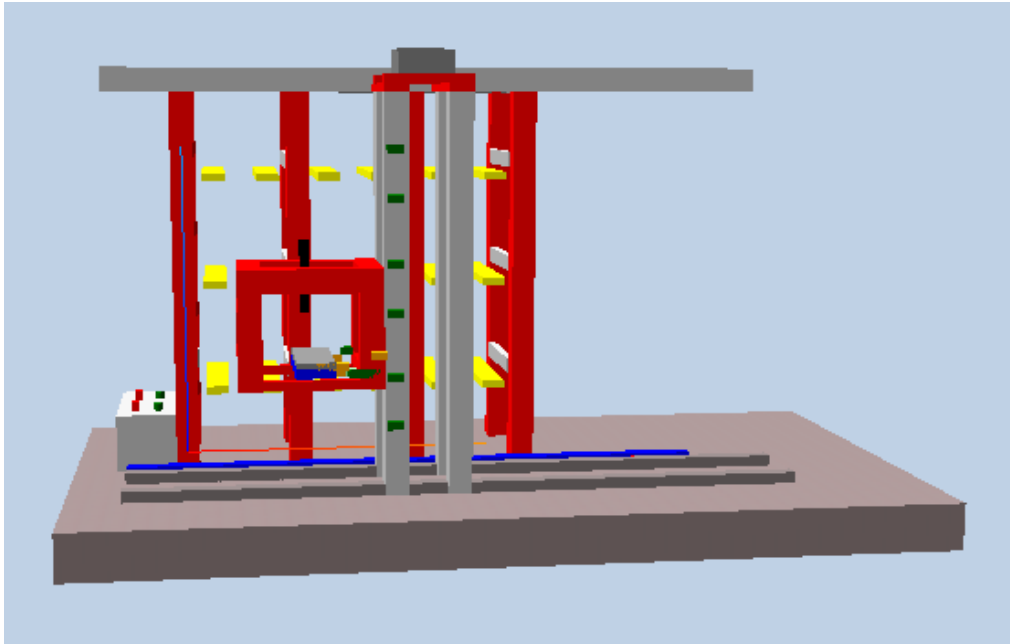


C:\str\labwork1\Debug\labwork1.exe

Starting web server on port 8081

LABWORK1 PROJECT: Example #2

- ❑ Try another example
- ❑ Use the code on the right into your source code file and try it... what's happening? How do we model the simulator movements?



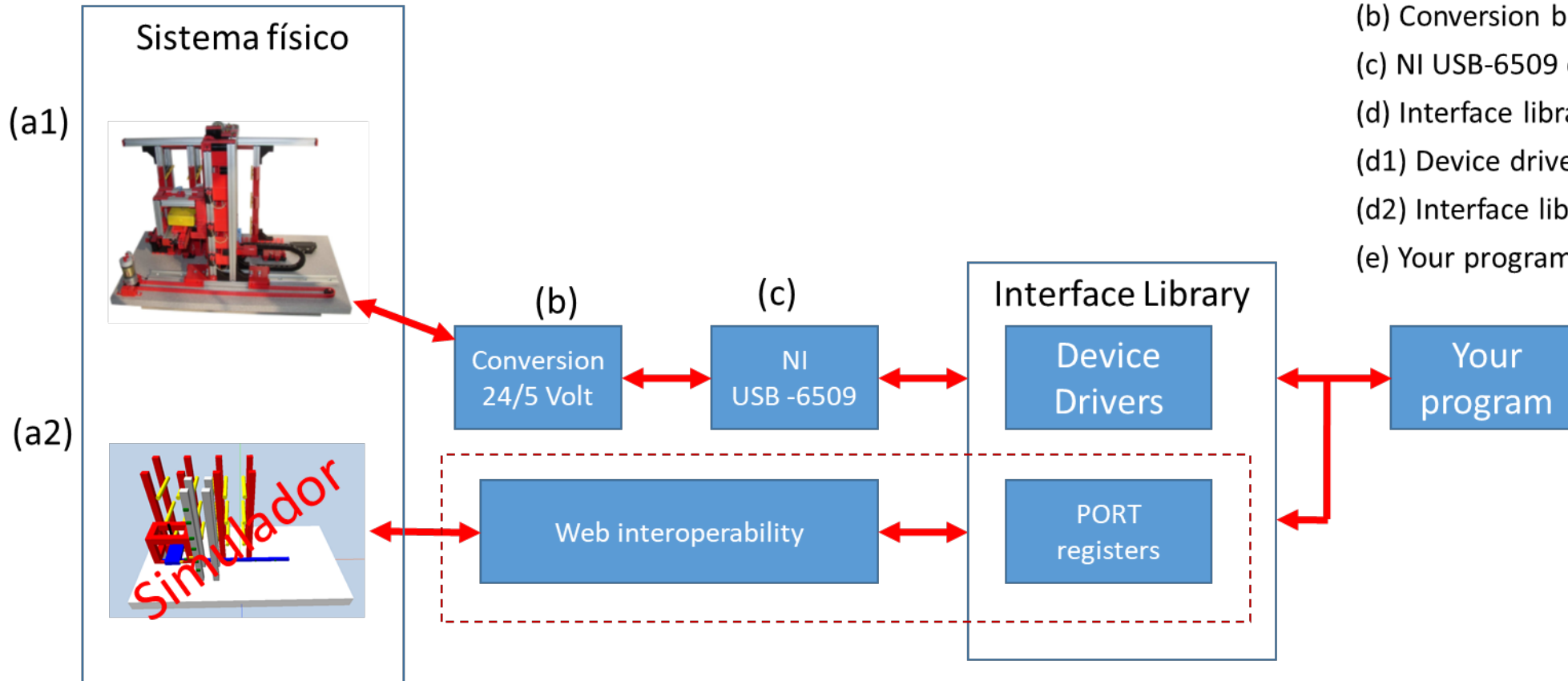
```
#include<conio.h>
#include<stdlib.h>
#include <windows.h> //for Sleep function
extern "C" {
    #include <interface.h>
}

int main(int argc, char **argv) {
    createDigitalInput (0);
    createDigitalInput (1);
    createDigitalOutput(2);
    uInt8 p0 = 0;
    uInt8 p2 = 0;
    writeDigitalU8(2, 0x80); // start moving right
    while (!_kbhit()) { //do until a key is pressed
        p0 = readDigitalU8(0);
        p2 = readDigitalU8(2);
        if (!(p0 & (1 << 0))) { // reached right side?
            p2 &= ~(1 << 7);
            p2 |= (1 << 6);
            writeDigitalU8(2, p2); // move left
        }
        if (!(p0 & (1 << 2))) { // reached left side?
            p2 &= ~(1 << 6);
            p2 |= (1 << 7);
            writeDigitalU8(2, p2); //move right
        }
    }
    writeDigitalU8(2, 0);
    closeChannels();
}
```

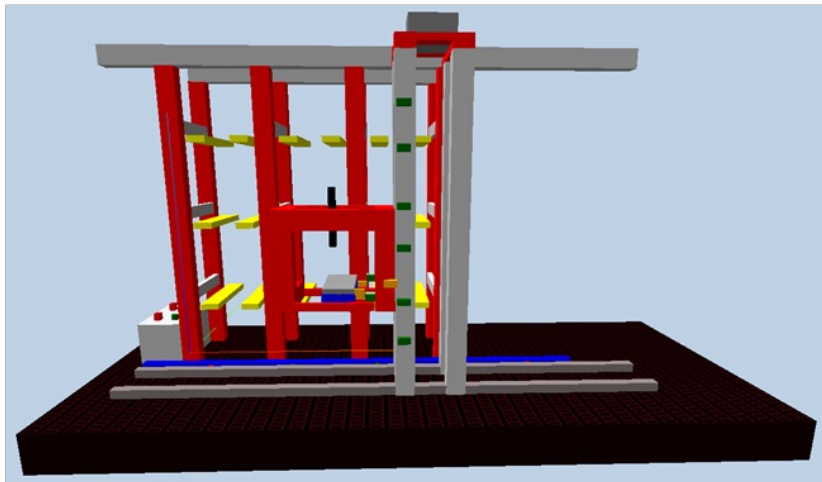
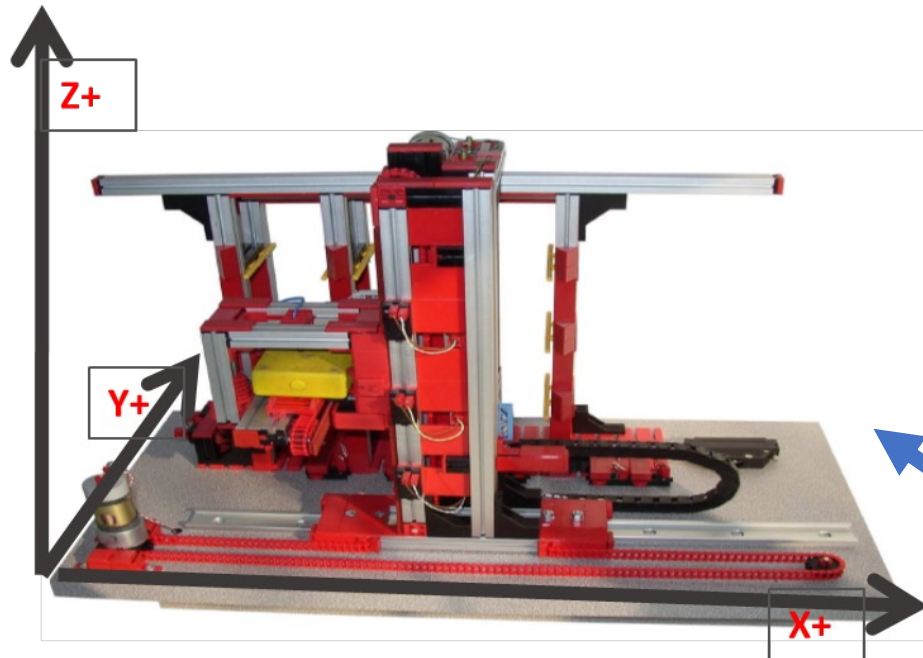
LABWORK1 PROJECT: Architecture Overview

- Switching between simulator/versus physical storage kit

- (a1) Real system
- (a2) Corresponding simulator
- (b) Conversion between 24/5 Volt
- (c) NI USB-6509 digital acquisition board
- (d) Interface library (supplied)
 - (d1) Device drivers
 - (d2) Interface library
- (e) Your program



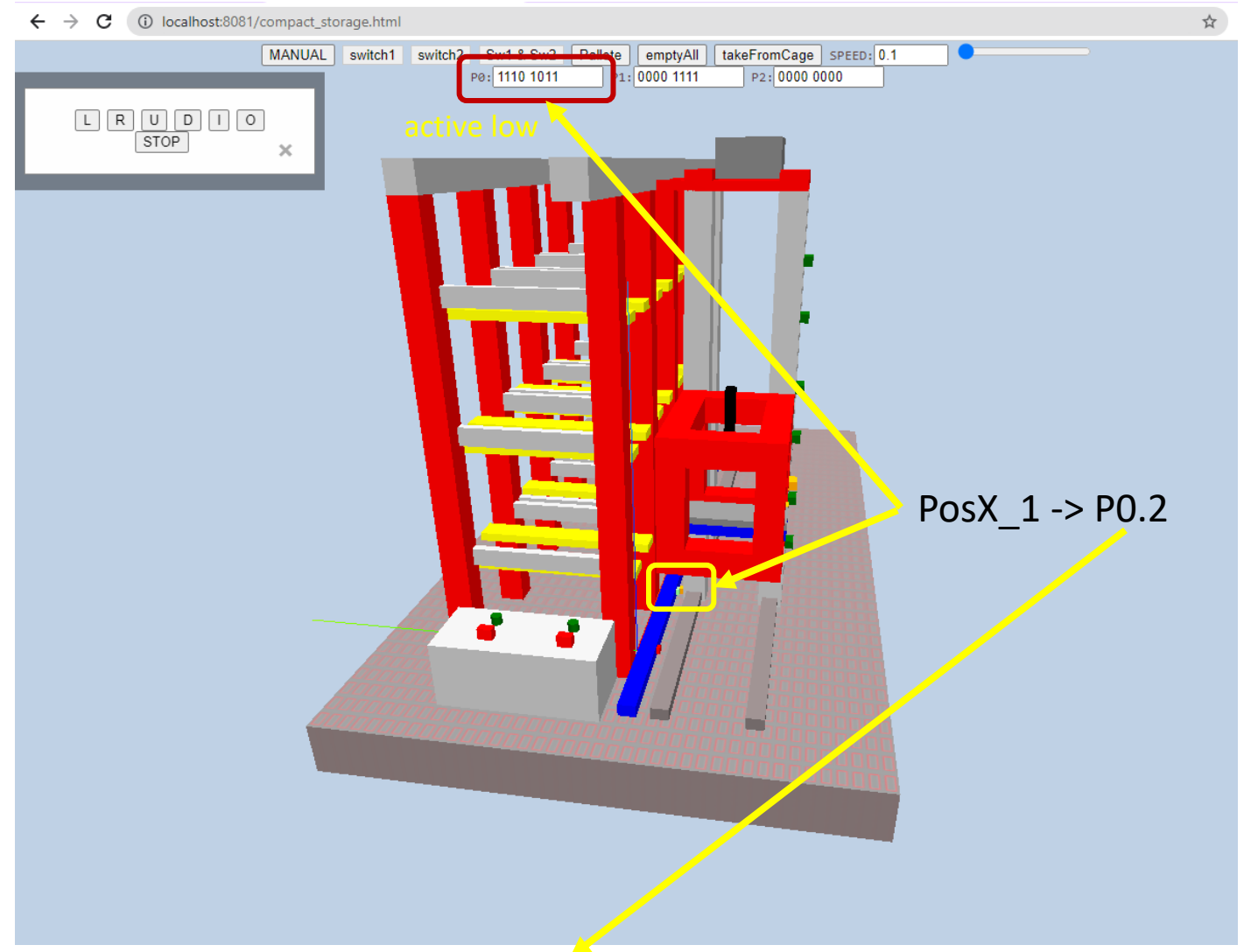
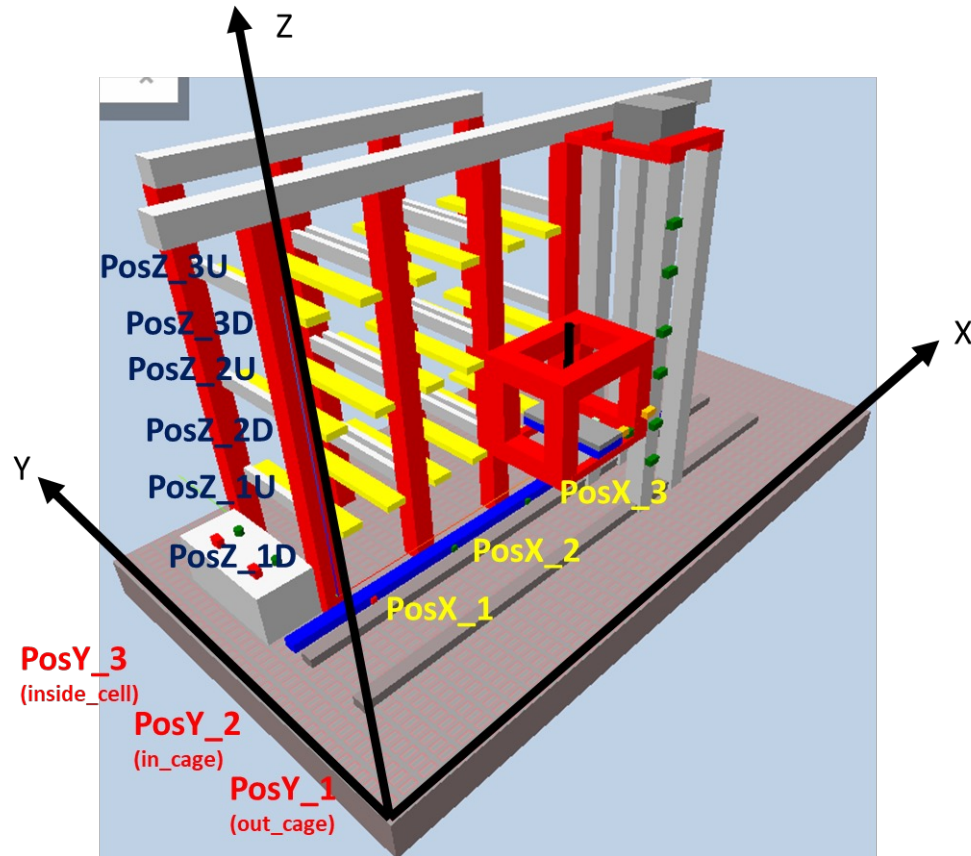
LABWORK1 PROJECT: Switching between kit & simulator



```
sim_server.c  interface.c  labwork1.cpp
labwork1 (Global Scope)
1  // interface.c
2  // J. Rosas
3
4  #include <stdlib.h>
5  #include <stdio.h>
6  #include <conio.h>
7  #include <signal.h>
8  #include <winsock.h>
9  #include <interface.h>
10
11  // #define _USE_REAL_KIT_
12
13  #define MG_HTTP_FOLDER "c:/str/simulators"
14  #define MG_HTTP_PORT  "8081"
15
```

LABWORK1 PROJECT: Work Assignment

- Considering the following positions



- Determine the tuple $P\langle \text{port.bit} \rangle$ for each position

LABWORK1 PROJECT: Work Assignment

□ Filling in the table below:

Axis	Position	Tuple p<port.bit>	OBS.
X	PosX_1	P0.2	active low
	PosX_2	P0.1	
	PosX_3	P0.0	
Y	PosY_1	...	
	PosY_2	...	
	PosY_3	...	
Z	...	...	
	...	...	

☐ And the actuators??

Move right
Move left
Move inside
Move outside
...



☐ Complete the table!



KEEP UP THE
GOOD
WORK!